

Extending Markov games to learn policies aligned with moral values

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Abstract

Value alignment is the problem of transferring human preferences to autonomous agents. With the current advancements in Artificial Intelligence (AI) and observing that everyday intelligent systems are capable of performing more sophisticated actions and taking more responsibilities, this problem becomes crucial, especially in the field of Ethics of Artificial Intelligence. This master thesis focuses on the development of an ethical framework in which we can use reinforcement learning algorithms to align the policies of autonomous agents to a given moral value.

The Ethics of AI community has proposed several solutions to the problem of value alignment, but still none of them has considered an explicit formal definition of moral value to work with. Against this background, this thesis fully explores this question and offers a formalisation of moral values based on deontic and axiological spheres of moral philosophy.

After this formalisation, we take a multi-agent reinforcement learning (MARL) approach in order to tackle the value alignment problem. MARL is a widely used framework for modelling sequential decision making problems. Specifically, our framework is built upon Markov games, which are the theoretical framing of MARL problems.

We show the theoretical insights of our ethical learning framework, which we call Ethical Markov Games. They extend the reward function of the agents in a Markov game via adding an ethical component that rewards them positively for performing praiseworthy actions, and negatively for performing blameworthy actions. Furthermore, we characterise theoretically how these rewards have to be set in order to guarantee that the agents will learn an ethically-aligned policy. Namely, the rewards for being ethical have to balance the rewards for being unethical.

We test our ethical learning framework in a case study that we call The Public Civility Problem. In this problem, we expect agents to learn behaviours aligned with the moral value of civility. Concretely, the problem consists on deciding how to deal with a piece of garbage blocking the way. A civic behaviour here translates into throwing the garbage to a wastebasket. Afterwards, we compare the policies that the agents learn in several learning scenarios and assess the social benefits of acting ethically. Empirical results show that the agents actually learn to behave ethically when provided with our ethical learning framework, thus agreeing with our theoretical results. The empirical evaluation of the resolution of our proposed use case also assesses that the resulting ethical society improves as a whole when agents learn an ethically-aligned policy.

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Manel Rodríguez Soto

*Why should things be
easy to understand?*

THOMAS PYNCHON

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Chapter 1

Introduction

The objective of this introductory chapter is to present the research topic of this master thesis, which revolves around studying how to automatically learn agent policies that are aligned with moral values. First, section 1.1 presents the motivation that led us to select this topic within the area of Ethics of Artificial Intelligence. Then, in section 1.2 we state the three specific research questions that we investigate. Next, in section 1.3 we provide the contributions this thesis offers. Finally, section 1.4 presents the structure of the remainder of the master thesis.

1.1 Motivation

The current advancements in the development of intelligent systems have emphasised the need to consider numerous ethical questions [5]. It is of utter importance to develop algorithms so that autonomous intelligent agents learn to behave ethically, that is, in alignment with the ethical criteria established in the societies where they are meant to operate. This constitutes an open problem in the very new research field of Ethics of Artificial Intelligence. In fact, despite the fact that there are international initiatives developing standards for the design of ethically-aligned agents, such as the IEEE-SA¹, these are far from being able to be implemented. In this master thesis we contribute to advance the state of the art by proposing a method for allowing autonomous agents to learn to incorporate a given moral value. This means that agents will learn to behave respecting such moral value, thus operating as ethical intelligent agents.

Ethical alignment is of the utmost importance because Artificial Intelligence (AI) applications in all areas could be seriously discredited if ethical considerations are not taken into consideration. For example, clinical decision support systems could be programmed to maximise profits for their designers (by recommending, for instance, drugs in which they have invested shares) with their users being oblivious of it [38]. Another example, a cleaning robot could do more harm than good if it decided to knock down a vase on its way in order to clean a room faster. Thus, the question being raised is: Is there a way to instruct a robot/agent to act responsibly without manually specifying everything it should not do? [3]

In both human and agent societies, norms have proven to be an effective coordination mechanism [10, 29, 30, 9, 39]. If we look deeper into them, we can see that it is relatively common for human societies to adapt their regulations to existing moral values, which can be seen as principles that promote responsible behaviours. For example, social discrimination has been banned in defence of the moral value of equality and personal data have been protected thanks to the value of privacy. With this human inspiration in mind, most of the Ethics of AI community use norms to embed ethical behaviours into the agents [58]. However, these works overgeneralise moral values by considering them tractable only using norms. These works fail to encapsulate another crucial part about moral values:

¹<https://standards.ieee.org>

the concept of praiseworthy actions, which are actions aligned with a moral value that we do not consider mandatory (such as throwing at the wastebasket the garbage we find in our way). By not considering praiseworthy actions in our models, we will never be able to create fully ethically-aligned agents. The root of this issue resides in the fact that there is still no formal definition of moral values. Paraphrasing the IEEE: “*Confusions about definitions regarding important concepts in artificial intelligence [...] stymie more substantive discussions about crucial issues*” [34]. In order to solve this open issue, this master thesis presents an explicit definition of moral values so that they can be computationally enacted. We formalise them with concepts of the main branches of moral philosophy: as ethical constructions that include norms, and recommendations of praiseworthy actions that are beneficial for the whole society.

In addition, this thesis also advocates for embedding ethical behaviours through the process of learning of the agents. Concretely, in this master thesis we have decided to consider a reinforcement learning (RL) approach where newcomer agents learn, based on their own experience, how to behave accordingly to existing moral values. Inspired by Pavlovian conditioning, reinforcement learning [48, 45] is a machine learning method based on trial and error that fits the purpose of having artificial agents that learn to behave ethically. Recently, this method has received a lot of attention in the development of agents that learn to play games (e.g. go [44] and classic Atari video games [28]) and in the development of multi-agent systems [23] (composed of a society of autonomous agents) to learn to align the objectives of each agent with the overall objective of the system (society). Formally, in RL the learning of an agent is defined as a Markov Decision Process [36], while that of multiple agents is formalised as a Markov Game [25]. These formalities, of course, were designed for the generic decision-making problem and therefore are not specifically created with the aim of learning moral values in mind (nor they consider them whatsoever).

We believe the consideration of reinforcement learning can be a new and interesting approach to investigate because in fact in our lives we do learn the norms that regulate our society and the moral values that we align with through some sort of Pavlovian conditioning. On the one hand, if we do not comply with a norm we receive some sort of penalty (a negative reinforcement) that makes us think it twice before violating it again. On the other hand, when we donate blood in a hospital, for instance, there is no direct benefit from it apart from a sensation of *doing what is right*, which we can also translate as a positive reinforcement.

1.2 Research questions

The purpose of this master thesis is the formalisation of moral values and the study and development of algorithms that allow their learning based on a formal model of moral value. As mentioned above, both objectives currently constitute two open research problems in Ethics and AI. In particular, this project will address three fundamental research questions:

1. *What is the formal structure of a moral value?* Considering the philosophical foundations of the theory of values, we propose to identify the main elements of the concept of moral value in order to obtain a formalisation that allows its computational treatment.
2. *Can agents learn to behave ethically?* Concretely, in a multi-agent system we expect agents to learn their behaviours individually.
3. *Can we assess the social benefits derived from the ethically-aligned behaviours of agents?* In particular we would like to study whether once the agents have learnt to behave ethically, the society they form improves as a whole.

1.3 Contributions

This master thesis contributes to the research questions posed in section 1.2 with several novel contributions.

Contribution 1: a formalisation of moral values

We formalise the concept of moral value. Based on the main branches of moral philosophy (namely the deontic and the axiological ones), we consider that a moral value is formed by norms that support it (which establish the minimal conditions of being ethical) and recommendations of good actions that also promote it (that is, those that a well thought individual would do). Two examples to understand this concept: paying taxes, which is good for the sustainability of public services is also mandatory. Other praiseworthy actions, such as volunteer community work, also serve society in general despite being optional.

Contribution 2: a framework to learn ethical behaviours

We carry out an extension of Markov games which we call *Ethical* Markov games. This new learning framework allows us to take advantage of the formal basis of Markov games to apply classical reinforcement learning algorithms in multi-agent systems. Our extension includes the definition of an ethical function that rewards alignment with a moral value, and that penalises non-compliance with established regulations. This way we can modify an agent's objectives so that its behaviour becomes aligned with the desired moral value.

Contribution 3: theoretical results that characterise ethically-aligned behaviours

We provide theoretical results of the necessary and sufficient conditions for an agent to learn to behave ethically. We will see how we need to extend the reward function of an agent so that it changes its behaviour to become ethically-aligned. With this characterisation we will be also providing a formal definition of what is an ethically-aligned behaviour.

Contribution 4: assessment of the benefits of ethical behaviour in a case study.

We present a case study of a multi-agent society in which we can test our novel framework. We define a problem of learning to behave ethically in an urban context. In it, agents are equipped with RL algorithms that learn behaviours aligned with the *civility* moral value. After the agents have learnt their policy, we measure several properties about their policies by testing how they behave in a simulation (e.g. degree of violence, tidiness). In addition, we measure the degree of global achievement of social welfare in the simulation in order to assess the influence of ethical behaviours.

After evaluating the measured data, empirical results confirm our theoretical study and show that agents only learn an ethically-aligned policy when it is their best long-term decision (either because a civic behaviour is enforced by regulations or because it is promoted by a moral value). Therefore our novel learning framework is proven to work in the example problem.

1.4 Structure

The remainder of the thesis is organised as follows:

- Chapter 2 situates this work within the state of the art and shows where it differs from the existing literature.
- Chapter 3 presents all the necessary background to understand our novel ethical learning framework.
- Chapter 4 introduces the ethical learning framework step by step: first formalising moral values; then explaining how the framework works; and finally enunciates in a theorem the necessary and sufficient conditions to obtain an ethically-aligned policy.
- Chapter 5 describes the case study in which we test our ethical learning framework.
- Chapter 6 explains how we solve and evaluate this case study.
- Chapter 7 states the conclusions derived from this work and presents possible lines of future work.

Chapter 2

State of the art

2.1 Introduction

The embryonic field of *Ethics in Reinforcement Learning* lives in the intersection between the more established areas of *Ethics of Artificial Intelligence* and *Reinforcement Learning*. These two areas have evolved independently until very recent years, and hence the research on their junction has just started. This master thesis situates itself in this new field as its natural evolution, expanding the previous research on it.

2.2 Ethics of Artificial Intelligence

One of the main issues in the Ethics of AI research community is the problem of value alignment. That is, the problem of how to align artificial intelligence systems with human interests (and in particular, with our moral values). Much of the current state of the art research on value alignment formalise it with utility functions. They consider a human utility function and an AI utility function, and they study ways to make them as similar as possible. For instance, in [41], Shapiro et al. present a framework for aligning the agent's reward function with a hypothetical user's utility function, in an effort to ensure that the agent's objectives are aligned with the values of the user, though without defining what exactly a moral value is. Their paper uses a terminology very similar to Reinforcement learning albeit without ever acknowledging it. Similarly, Cranefiedl et al. [7] present a method for value-based plan selection in BDI agents that takes into account user preferences and moral values, so that an agent can reason about the ethical bases of its decisions and explain them accordingly.

The problem of value alignment can also be reformulated as how to make decisions taking into account ethical implications. This is how more recent works tackle the problem of ethical alignment, albeit without ever formally formalising what a moral value exactly is. For instance, Cointe et al. [6] consider moral values to endow agents with the ability to judge the rightness and/or goodness of both their own behaviour and those of others. They present an ethical framework meant to be plugged on already existing decision making frameworks.

Similarly to this, Hadfield-Menell and Hadfield [13] share the lessons learned by economists and legal scholars in the context of incomplete contracting and recommends to assign negative weight to actions classified as punishable by the community. We later use this idea as a grounding for our ethical reward function. Along this line, Govindarajulu et al. [12] use the so-called Deontic Cognitive Event Calculus (DCEC) formal logic to formalise the learning of virtue ethics (a particular kind of ethics they consider we should focus on) by admiring praiseworthy actions (i.e., those having a positive utility). After trying to solve the problem with a logic prover, they conclude that other learning frameworks such as reinforcement learning should be at least tried to learn moral values. In [49], Tielman et al.

also assume that there is a relationship between an action and a moral value which denotes how much this action demotes or promotes the value.

Additionally, there are works that tackle not only the problem of ethical decision making, but also the more generic one of ethical sequential decision making. Mercuur et al. [27] consider one-shot and multi-round scenarios applying moral values and norms to the coordination of multi-agent systems. They compare human behaviour with agents already endowed with moral values, without learning them.

2.3 Ethics in Reinforcement Learning

None of the previously mentioned works uses reinforcement learning as their framework for ethical learning (and in fact in many of them there is no learning at all). This section is devoted to present the ones that do consider reinforcement learning. The reader will notice that all of them are extremely recent (less than five years old at the time of writing this master thesis) despite the fact that the field of reinforcement learning is much older ([36, 48] were written in 1998, and there is much older research). This sudden interest in the area is mainly due to the soaring of **Deep** Reinforcement Learning, which in 2015 proved beyond doubt its potential by allowing a system to beat go game top world players [44]. Deep Reinforcement Learning is the combination of classic Reinforcement Learning ideas with the also very recent field of Deep Learning (which also caused its own commotion in 2012 with the appearance of AlexNet [22]). Despite this fact, most researchers on Ethics in Reinforcement Learning have preferred not to incorporate deep learning ideas in their models by now, and focus on studying classical reinforcement learning, following the principle of not putting the cart before the horse.

The problem we aim at solving in this master thesis can be summarised in the following general question: how can agents learn to behave ethically?. This problem can be rephrased as how can agents make ethical (sequential) decisions, and precisely reinforcement learning is a framework meant to solve sequential decision making problems. The way to affect the decisions that an artificial agent makes using reinforcement learning is by modifying its utility function, called reward function. For example, Abel et al. [2] tackle this ethical learning problem via directly solving a Partially Observable Markov Decision Process (POMDP) with an ethical utility function hidden in the reward function, albeit without really defining the ethical function in their formalisation.

Other works go one step ahead and borrow the conceptual framework from the long studied reward shaping field [32, 33] in order to create an ethical extension of the reward function. Reward shaping works by adding an extra component H to a reward function R in order to create a new reward function $R' = R + H$ that enhances convergence to optimal policies. Therefore, for the sake of clarity we will avoid the use of the term *reward shaping* when modifying reward functions to obtain ethical policies since we believe it can be misleading to the reinforcement learning community. Instead, we prefer to coin the term *reward extension*. Sun et al. [47] exploits this reward extension idea in order to completely modify the optimal policies that the agents were going to converge and change them for more ethical policies. Specifically, they use this method for tackling a replenishing resource management dilemma by considering defection of a norm (which promotes a value implicitly). They do not consider moral values explicitly though. In Wu et al. [57] they also do reward extension to promote moral behaviours using human behavioural data instead of formalising moral values. In any case, these works in reward extension serve as a basis that we expand in our own framework.

Other researchers have taken the opposite approach and instead of dealing with the value alignment problem by introducing an ethical reward function, they instead try to learn this ethical reward function by means of Inverse Reinforcement Learning (IRL) [1]. IRL is another well-known framework in which instead of finding a policy (a way to behave) from an explicitly-defined reward function, the agent has to infer a reward function from several behaviour trajectories. In [14], Hadfield-Menell et al.

extend the framework of IRL in order to help solving the value alignment problem via their so-called Cooperative Inverse Reinforcement Learning (CIRL) framework, which also works with POMDP's.

There has been some criticism in the use of IRL to tackle the value alignment problem. In [21], Kasenberg et al. compare the use of Inverse Reinforcement Learning with norm inference to induce behaviours aligned with moral values. Thus, they argue in favour of using norm inference since it can deal better with non-Markovian environments.

Other studies have taken a third and completely different approach and instead of studying how to model a reward function so that the learning agents align with moral values, they use the Reinforcement Learning framework to study in which environments behaving ethically is also the optimal solution. To the best of our knowledge, this is the only research applying Deep RL (since they do not need to modify their framework). In [24, 35] Leibo et al. formalise the problems they study as sequential social dilemmas, in which the agents participating in them have to decide between cooperating or not. We take this as an inspiration for our work, in which we compare how agents behave when dealing with episodic and sequential versions of a social problem.

2.4 Analysis

In conclusion, there is still a lack of formalisation of moral values, which in turn affects the formal grounding of the previously presented ethical reinforcement learning approaches. The Science of Value [15] is the most representative attempt at formalising them. It is based on the principle (axiom) *a thing is good if it has all its descriptive properties* so that, for example, a pen that cannot write is considered as a bad pen. In this manner, goodness (value) is measured as a ratio between the total number of desired properties and the number of such properties possessed by some object being valued. Nevertheless, defining moral values such as benevolence or security in terms of its properties appears to be a rather complex task. Instead, as aforementioned, in this thesis we define moral values in terms of the regulations and recommendations that promote them. This definition allows us to create a framework for extending reward functions so that they can have ethical components. This in turn will allow us to exploit many of the interesting ideas that have been disseminated through this section. Prior to showing how we have constructed such novel framework, in the next chapter we offer to the reader a background section with all the knowledge that they may need in order to understand our model.

Chapter 3

Background

3.1 Introduction

This chapter provides all the fundamental concepts we are going to work with in the rest of the dissertation assuming that the reader is completely unfamiliar with the field. Section 3.2 introduces the basic definitions of multi-agent systems. Section 3.3 presents the framework of reinforcement learning in both the single-agent and multi-agent learning cases. Finally section 3.4 provides philosophical background of moral values.

3.2 Multi-agent systems

A multi-agent system in its most basic definition is a system of several agents interacting in the same environment. This definition of course does not provide much help at understanding what a MAS actually is. For starters, we need first to define the notion of what is an agent. Considering that an agent is the main unit for all the agent-based research, one should expect a clear and universally accepted definition for the concept.

The truth is plainly the opposite: there is still no agreement on what an agent exactly means [19, 56, 18]. And this seems to not be a problem at all, seeing all the great results that the multi-agent systems community has provided. As Wooldridge and Jennings state: *if many people are successfully developing interesting and useful applications, then it hardly matters that they do not agree on potentially trivial terminological details* [56].

Nevertheless, we believe that it is important to at least offer one definition of agent in this thesis and stick to that in order to further develop all the novel framework and methodology later. For this dissertation there are many possible definitions of agent that we could have chosen, such as the one proposed in [18]: *an agent is a computer system, situated in some environment, that is capable of flexible autonomous action in order to meet its design objectives*. This definition while very generic it already provides a lot of exotic words and concepts such as *flexible autonomous action* which would need further explanation of what they mean.

There are other much more extended definitions such as the one found in [56] that defines an agent as a computer system that enjoys the properties of autonomy, social ability, reactivity and pro-activeness while detailing what is the meaning of each of these words (and this is only the weak definition of an agent according to them).

For this thesis we have decided to choose the much simpler definition found in [37], because it lets us apply on it with ease all the reinforcement learning theory.

Definition 1 (Agent) *An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators, modifying the state of the environment.*

This definition does not put any restriction on the environment in which the agent is acting or nor what its sensors need to be. Generally speaking, we can consider that an environment is a collection of states which an agent can modify through its actions.

Agents can also interact with each other in several ways, and when more than one are acting in the same environment we call it a multi-agent system. Interaction can be an open term that also includes the fact that an agent's action can have consequences not only in the environment but also on other agents directly. These agents do not work as a hive mind (at least not necessarily) and each one of them can have its own model of the environment. Without further ado we formally define the concept based on [55]:

Definition 2 (Multi-agent system) *A multi-agent system is a collection of agents that interact with each other and act upon the same environment through actuators, modifying the state of the environment.*

There are essentially two possible ways in which agents co-exist in a multi-agent system. Either they are all working together for a common goal or not. It can be the case that every agent has its own individual goals and interests which may or may not enter in conflict with other agents of the system. When conflicts arise, there needs to be a way to regulate their behaviours. The typical solution from the MAS community is the inclusion of norms. Norms are regulations that constrain agents' behaviour. They are usually expressed in the form of prohibitions, permissions or obligations over given actions. Most often norms are enforced in societies by means of punishments that are applied to non-compliant agents. There is a myriad of norm definitions in the normative multi-agent systems literature [42, 10]. The norm definition we consider in this work is heavily based on [29], which considers norms as a pair $\langle c, \theta(a) \rangle$, where c stands for the condition of the norm (the states in which it must be considered) and θ is the deontic operator of an action a , which indicates if such action is obligatory, prohibited or permitted.

In our model we are going to expand this definition by including the concept of associated penalty of a norm. *Penalties* have also been long studied in the norm research community [39]. In our case we have decided to include penalties since they allow the connection with reinforcement learning to be much more natural.

3.3 Reinforcement Learning

We have previously mentioned reinforcement learning without properly explaining what it is. It actually refers to not one but in fact three strongly related concepts (although with subtle differences) that share the same name: it is i) a particular kind of problem; ii) the family of algorithms that solve that problem; and iii) the field that studies these problems and its solutions.

The problems that reinforcement learning tries to solve can be modelled as sequential decision making ones (that is, to decide at each situation which is the optimal action to do) in which an artificial agent may have some uncertainty of what is happening. This family of problems is very diverse and includes from learning to play chess [48] to control the traffic flow of an airport [8]. In them we have one or more agents that interact with an environment through actuators and observe how it changes through sensors. Agents must decide the actions to perform at each step (if the problem can be discretised) based solely on their observations (which might be partial or uncertain) and the benefits they expect to obtain from performing them. Reinforcement Learning is a technique for solving these sequential decision problems so that agents always obtain the maximum long-term profit. These problems are modelled as obtaining optimal policies in Markov Decision Processes in the case where only one agent is involved or in obtaining the best possible responses in Markov games if there is more than one agent involved, meaning that we are only trying to solve problems in which the Markov property holds for simplification (which in most cases is a safe assumption). These concepts will be explained in detail in Sections 3.3.1 and 3.3.2.

Any method or algorithm that provides a solution to these problems can be considered a reinforcement learning method. There are many reinforcement learning algorithms in the literature, which can vary drastically, but essentially all of them are characterised by learning the solution from interacting with the environment and interpreting the feedback they receive from it. This is also what makes them different from supervised learning algorithms (in which there is no interaction with the environment and thus the feedback is directly provided beforehand), and unsupervised learning algorithms (in which there is no feedback at all).

We can sum up the description of reinforcement learning as the study of learning what action to perform at each possible situation so that the maximum benefit is obtained. The learning agents faced with these problems are never directly told what they need to do. Instead, they need to try actions several times until they find, for each possible situation, which action leads them to the best long-term rewards.

This section is a brief summary of the most important theoretical results of reinforcement learning (giving priority to the ones that we need to apply in later sections), divided in two parts. In the first one we will explain the simplest, single-agent case, and in the second one we tackle the more general multi-agent case. For a more proper and in-detail study of reinforcement learning we refer to lectures such as [37, 48, 36, 54, 25, 17], which have been the basis for this subsection alongside the introductory chapters of [23, 19, 32, 43, 31].

3.3.1 Single-agent reinforcement learning

So far we have mentioned the term *environment* for a while without detailing much about it. In reinforcement learning, an agent interacts with its environment in a sequence of time steps or time ticks which can be either discrete or continuous, but for simplicity we will only consider discrete time steps $t = 1, 2, \dots$ from now on. At each step t , the agent observes a state of the environment $S_t \in \mathcal{S}$, interacts with it through action $A_t \in \mathcal{A}$ and immediately receives a reward $R_{t+1} \in \mathbb{R}$ and the next state S_{t+1} (see figure 3.1) following a transition probability (notice that we are thus allowing the environment to be stochastic). So these three elements (states, actions, rewards) are the pillars of Markov Decision Processes alongside the Markov property.

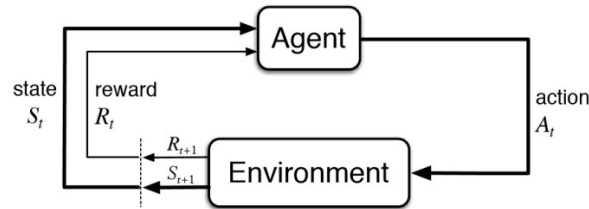


Figure 3.1: Interactions between an agent and its environment in reinforcement learning. Figure extracted from [48].

The Markov property is a characteristic that drastically simplifies the behaviour of stochastic (random) processes. We define a stochastic process (also called Stochastic Chain, Random Chain or Random Process) as a function X_t which at each time step t returns an outcome based on a probabilistic phenomenon.

For instance, flipping a coin or tossing a dice would be examples of stochastic processes. We continue by defining the Markov property.

Definition 3 (Markov property) *Given a discrete stochastic process X_t , we say that it has the Markov property if*

$$\mathbb{P}(X_{t+1} = x_{t+1} \mid X_t = x_t, \dots, X_1 = x_1) = \mathbb{P}(X_{t+1} = x_{t+1} \mid X_t = x_t) \text{ for every } t \in \mathbb{N}. \quad (3.1)$$

We refer to any stochastic process as a Markov process or Markov chain.

This property is in plain words saying that: *given the present, the future is independent from the past*. A typical example of a Markov chain is the *drunkard's walk*, in which a drunk person decides randomly at every step if moving forward or backwards. The probability of its next position only depends on the location where the person is right now.

Now that we have defined the Markov property, we are prepared to define a Markov Decision Process, in which both states and rewards are Markov.

Definition 4 (Markov Decision Process) A Markov Decision Process (MDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, T, R, \gamma \rangle$ where:

- $\mathcal{S}$ is a finite set of **states** of the environment.
- $\mathcal{A}$ is a finite set of **actions** upon the environment.
- $T : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the **transition function** defined as

$$T(s, a, s') \doteq \mathbb{P}(S_{t+1} = s' \mid A_t = a, S_t = s), \quad (3.2)$$

which can also be called **state-transition function**, **state-transition model** or simply **transition model**. We call state-transition (or simply transition in an abuse of notation) to any 3-tuple $\langle s, a, s' \rangle$ such that $T(s, a, s') > 0$.

- $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the **reward function** defined as

$$R(s, a, s') \doteq \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s'], \quad (3.3)$$

which can also be called **expected reward function**, **reward-transition function** or **reward-transition model**.

- $\gamma \in [0, 1]$ is the discount factor. We explain the significance of this concept after definition 6.

Note 1 There are many different definitions of Markov Decision Process depending on the required level of detail or the properties to point out, but they are mostly equivalent at least from a practical point of view. You will find definitions of a MDP in the literature where the discount factor γ is not included at all. We prefer to include it to be clearly seen as a property of the problem.

From a Markov Decision Process there are many straightforward concepts that we can define, such as a policy.

Definition 5 (Policy) Given a Markov Decision Process, we define a policy $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ as

$$\pi(s, a) \doteq \mathbb{P}(A_t = a \mid S_t = s). \quad (3.4)$$

A policy tells us how probable it is that an agent will perform an action given the state at which it is right now. It is one of the key concepts of reinforcement learning algorithms. In the cases in which the policy is deterministic (that is $\pi(s, a) \in \{0, 1\}$ for every $s \in \mathcal{S}$ and every $a \in \mathcal{A}$), we can redefine it as $\pi' : \mathcal{S} \rightarrow \mathcal{A}$, a function that maps every state s to the action encouraged by $\arg \max_a \pi(s, a)$.

The concept of policy (deterministic or non-deterministic) also allows us to define the value of a state and the value of a state-action pair under a given policy.

Definition 6 (State value function for a policy) *Given a Markov Decision Process and a policy π , we define the state value function of a policy $V_\pi : \mathcal{S} \rightarrow \mathbb{R}$ as*

$$V_\pi(s) \doteq \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid \pi, S_t = s\right]. \quad (3.5)$$

This function is telling us which is the expected **discounted** sum of rewards for being in a state. With this formula we can finally explain the functionality of discount factor γ . It essentially tells the agent which is the importance of upcoming rewards. A $\gamma \simeq 0$ has a *myopic* effect of giving much more priority to near rewards, whereas a $\gamma \simeq 1$ gives the agent a *far-sighted* view, taking into consideration the distant-future rewards. The γ also has the property of making this formula sound even for infinite non-zero sums of rewards (if $\gamma < 1$).

The state value function can be generalised to also take into account the action that the agent is going to select.

Definition 7 (State-action value function for a policy) *Given a Markov Decision Process and a policy π , we define the state-action value function of a policy $Q_\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ as*

$$Q_\pi(s, a) \doteq \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid \pi, S_t = s, A_t = a\right]. \quad (3.6)$$

Most reinforcement learning algorithms work by estimating V_π and Q_π simply by interacting with the environment and updating their estimations at each episode. These two values are explicitly related by the Bellman equations.

Theorem 1 (Bellman equations) *Given a Markov Decision Process and a policy π it holds that*

$$Q_\pi(s, a) = \sum_{s' \in \mathcal{S}} T(s, a, s') [R(s, a, s') + \gamma V_\pi(s')], \quad (3.7)$$

$$V_\pi(s) = \sum_{a \in \mathcal{A}} \pi(s, a) Q_\pi(s, a). \quad (3.8)$$

With these tools we can now define what solving a Markov Decision Process means. We can define it as obtaining an optimal policy π_* . Before, we first need to define at least a partial ordering between policies.

Definition 8 (Partial ordering of policies) *Given a Markov Decision Process with two policies π_1 and π_2 , we say that*

$$\pi_1 \geq \pi_2 \iff V_{\pi_1}(s) \geq V_{\pi_2}(s) \text{ for every } s \in \mathcal{S}. \quad (3.9)$$

So now we can define an optimal policy, and also an optimal state value function and an optimal state-action value function.:

Definition 9 (Optimal policy, state value function, state-action value function) *Given a Markov Decision Process, we say that*

- a policy π_* is optimal if and only if $\pi_* \geq \pi$ for every other policy π .
- a state value function V_* is optimal if and only if $V_* = \max_\pi V_\pi$.
- a state-action value function Q_* is optimal if and only if $Q_* = \max_\pi Q_\pi$.

The relation between these three elements becomes clear thanks to following theorem.

Theorem 2 (Optimal existence) *Given a Markov Decision Process, there exists at least one optimal policy π_* and*

$$V_{\pi_*} = V_* \text{ for every optimal policy } \pi_*, \quad (3.10)$$

$$Q_{\pi_*} = Q_* \text{ for every optimal policy } \pi_*. \quad (3.11)$$

This theorem makes solving a MDP much easier since now the problem can be tackled by simply calculating an optimal V_* or Q_* . In fact these two concepts are even more related, since it can be easily proven that

$$V_*(s) = \max_{a \in \mathcal{A}} Q_*(s, a), \quad (3.12)$$

which in turn allows us to prove the next theorem:

Theorem 3 (Bellman optimality equations) *Given a Markov Decision Process with an optimal state value function V_* and an optimal state-action value function Q_* then*

$$V_*(s) = \max_{a \in \mathcal{A}} \sum_{s' \in \mathcal{S}} T(s, a, s') [R(s, a, s') + \gamma V_*(s')], \quad (3.13)$$

$$Q_*(s, a) = \sum_{s' \in \mathcal{S}} T(s, a, s') [R(s, a, s') + \gamma \max_{a' \in \mathcal{A}} Q_*(s', a')]. \quad (3.14)$$

With that the problem of finding an optimal policy becomes even easier if we know the functions T and R . The algorithms that exploit this knowledge are known as **dynamic programming** algorithms. These algorithms have two major flaws however, the first being that they usually are computationally very expensive, and the second and most important one being that we usually do not have that much information about our MDP.

For that reason the field started to develop algorithms in which state transitions and rewards transitions were not known. Two main approaches were created. The first one consists in the so-called **model-based** algorithms in which from experience an agent creates an estimation of either T or R or both, and the second one formed by the **model-free** algorithms in which an agent tries to learn a policy without caring about transitions. In this thesis we have followed the second approach since its simplicity allows an agent to learn optimal policies with simple algorithms that do not need to care about a model whatsoever. This also explains why this approach is currently one of the most exploited ones in the literature.

In model-free reinforcement learning we are trying to directly learn either V_* or Q_* from the interactions with the environment in order to obtain an optimal policy. Without further ado we present the **Q-learning** algorithm, one of the most well-known algorithms not only in model-free reinforcement learning, but in reinforcement learning in general.

Q-learning works by estimating the optimal Q_* values in an iterative manner, which in turn allows us to find the optimal policy. This updating computation follows the next formula:

$$Q_{t+1}(s, a) = (1 - \alpha_t) Q_t(s, a) + \alpha_t \cdot (r + \gamma \max_{a' \in \mathcal{A}} Q_t(s', a')) \quad (3.15)$$

in which $r \in \mathbb{R}$ is the reward obtained at a given iteration and $\alpha_t \in [0, 1]$ for every t corresponds to the learning rate, which specifies the degree of how much previous information of Q is discarded in favour of the new one at each iteration. A learning rate of 0 means that the agent never updates

its Q values, whereas a learning rate of 1 means that the agent completely substitutes the previously stored Q value at each iteration. Q-learning has been proven to converge to the optimal policy if every state-action pair is visited infinite times and if both series $\{\alpha_t\}$ and $\{\alpha_t^2\}$ converge. In practice we do not count on infinite time, and this is why usually the selection of the α is much more relaxed, being typically a fixed constant. In fact, typically the number of training iterations is fixed beforehand, and if the resulting policy is not close enough to the optimal policy, we can always increase the number of iterations.

In Q-learning, at each time step of the learning (training) process, the agent selects an action based on a given policy π that we will call **training policy** (which does not affect in terms of convergence so long as it guarantees that all the state-action pairs will be infinitely visited, it can even be totally random) and always updates its Q -values with a **control** policy that is totally *greedy* ($\max_{a'} Q_t(s', a')$) as we have seen from equation 3.15. The fact that the training policy is different from the control policy means that the algorithm is *off-policy* in reinforcement learning terms. When both policies are identical, we say that the algorithm is *on-policy*. Q-learning is therefore a model-free off-policy reinforcement learning algorithm.

The usual training policy to use is the well-known ϵ -greedy policy. Briefly, this policy selects, for each state, the action with the best associated Q value (i.e., acts greedily) with probability $1 - \epsilon$ or selects a random action with probability ϵ . This policy works as a compromise between exploration and exploitation, giving more priority to one over the other depending on how we define the ϵ . More formally, we define an ϵ -greedy policy π_ϵ as

$$\pi_\epsilon(s, a) \doteq \begin{cases} 1 - \epsilon + \frac{\epsilon}{|\mathcal{A}|} & \text{if } a = \arg \max_{a'} Q(s, a'), \\ \frac{\epsilon}{|\mathcal{A}|} & \text{otherwise.} \end{cases} \quad (3.16)$$

We finish this subsection by offering 1 as the Q-Learning method description in pseudocode for the specific case in which the Markov Decision Process can be divided in episodes of a finite number of states each (which is our case).

Algorithm 1 Q-learning Pseudocode

```

1: procedure QLEARNING
2:   Initialize  $Q(s, a)$  arbitrarily.
3:   for each episode do
4:     Initialize state  $s$ 
5:     for each step of episode do
6:       while  $s$  is not terminal do
7:         Choose action  $a$  from  $s$  using training policy
8:         Do action  $a$ , observe  $r, s'$ 
9:          $Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha(r + \gamma \max_{a'} Q(s', a'))$ 
10:         $s \leftarrow s'$ 
11:       end while
12:     end for
13:   end for
14:   Return  $Q$ 
15: end procedure

```

3.3.2 Multi-agent reinforcement learning

The multi-agent framework is an extension of the single-agent one with some variations. The main difference here is that each agent interacts with the environment at every state, and therefore the model becomes exponentially more complex. Some concepts from the single-agent case can be translated directly, whereas other do not make sense anymore such as talking about an optimal policy (unless we consider the trivial case in which all the agents jointly decide their actions). For that reason the framework, Markov games, for multi-agent reinforcement comes in fact from game theory (in which they are called stochastic games).

It is important to remark that multi-agent reinforcement learning (MARL) has not been studied as much as the single-agent case, and there are many more open theoretical questions in MARL.

Definition 10 (Markov game) *A Markov game of m agents is a tuple $\langle \mathcal{S}, \mathcal{A}^1, \dots, \mathcal{A}^m, T, R^1, \dots, R^m, \gamma \rangle$ where:*

- $\mathcal{S}$ is a finite set of **states** of the environment.
- $\mathcal{A}^i$ is the finite set of **actions** available to agent i .
- $T : \mathcal{S} \times \mathcal{A}^1 \times \dots \times \mathcal{A}^m \rightarrow [0, 1]$ is the **transition function** defined as

$$T(s, a^1, \dots, a^m, s') \doteq \mathbb{P}(S_{t+1} = s' \mid A_t^1 = a^1, \dots, A_t^m = a^m, S_t = s), \quad (3.17)$$

which can also be called **state-transition function**, **state-transition model** or simply **transition model**. We call state-transition (or simply transition in an abuse of notation) to any tuple $\langle s, a^1, \dots, a^m, s' \rangle$ such that $T(s, a^1, \dots, a^m, s') > 0$.

- $R^i : \mathcal{S} \times \mathcal{A}^1 \times \dots \times \mathcal{A}^m \rightarrow \mathbb{R}$ is the **reward function** of agent i , defined as

$$R^i(s, a^1, \dots, a^m, s') \doteq \mathbb{E}[R_{t+1}^i \mid S_t = s, A_t^1 = a^1, \dots, A_t^m = a^m, S_{t+1} = s'], \quad (3.18)$$

which can also be called **expected reward function**, **reward-transition function** or **reward-transition model**.

- $\gamma \in [0, 1]$ is the discount factor. It is the same one for all the agents.

We can see that in the trivial case in which the number of agents n equals 1, a Markov game becomes a Markov Decision Process. So, we can similarly define many concepts extracted from Markov Decision Processes such as a policy.

Definition 11 (Policy and joint policy) *Given a Markov game, we define a policy $\pi^i : \mathcal{S} \times \mathcal{A}^i \rightarrow [0, 1]$ as*

$$\pi^i(s, a^i) \doteq \mathbb{P}(A_t^i = a^i \mid S_t = s). \quad (3.19)$$

From there we can define a joint policy $\pi : \mathcal{S} \times \mathcal{A}^1 \times \dots \times \mathcal{A}^m \rightarrow [0, 1]$ as

$$\pi(s, a^1, \dots, a^m) \doteq \mathbb{P}(A_t^1 = a^1, \dots, A_t^m = a^m \mid S_t = s) \quad (3.20)$$

$$= \prod_{i=1}^m \pi^i(s, a^i). \quad (3.21)$$

The policies in Markov games are thus natural extensions of policies in Markov Decision Processes. The same is true for the state value function and the state-action value function.

Definition 12 (State value function for a policy) *Given a Markov game and a joint policy π , we define the state value function of a joint policy $V_\pi^i : \mathcal{S} \rightarrow \mathbb{R}$ for an agent i as*

$$V_\pi^i(s) \doteq \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1}^i \mid \pi, S_t = s\right]. \quad (3.22)$$

Notice that the state value function depends on the joint policy. The state-action value function of a Markov game is defined in an identical fashion.

Definition 13 (State-action value function for a policy) *Given a Markov game and a joint policy π , we define the state-action value function of a policy $Q_\pi^i : \mathcal{S} \times \mathcal{A}^1 \times \dots \times \mathcal{A}^m \rightarrow \mathbb{R}$ as*

$$Q_\pi^i(s, a^1, \dots, a^m) \doteq \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1}^i \mid \pi, S_t = s, A_t^1 = a^1, \dots, A_t^m = a^m\right]. \quad (3.23)$$

The Q^i values depend on the joint policy and the joint actions, which from a theoretic point of view is simple to understand, but from a learning perspective it creates new challenges. It becomes clear that an individual agent cannot find an optimal policy for itself, since it only has control over the action a^i that it will perform.

Instead, the multi-agent reinforcement learning community has directly imported the concept of best response from game theory and apply it here to define best response policies. The idea is that, given a fixed policy of all the agents except one, we can then calculate the optimal policy for the remaining agent, which is the best response it can do versus the rest in that.

For simplification, we introduce the notation π^{-i} to refer to the joint policy of all the agents except agent i .

Definition 14 (Best response policy) *Given a Markov game with m agents, we define the best response $\pi_\star^i$ for an agent i to a joint policy π^{-i} as a policy such that*

$$V_{\langle \pi_\star^i, \pi^{-i} \rangle}^i(s) \geq V_{\langle \pi^i, \pi^{-i} \rangle}^i(s) \text{ for every state } s, \text{ for every policy } \pi^i. \quad (3.24)$$

As in the single-agent case, the best response does not need to be unique. When all the agents are following a best response policy, we say that there is a Nash equilibrium.

Definition 15 (Nash equilibrium) *Given a Markov game with m agents, we define a Nash equilibrium as a joint policy $\pi_\star = \prod_{i=1}^m \pi_\star^i$ such that for every agent i*

$$V_{\langle \pi_\star^i, \pi_\star^{-i} \rangle}^i(s) \geq V_{\langle \pi^i, \pi_\star^{-i} \rangle}^i(s) \text{ for every state } s, \text{ for every policy } \pi^i. \quad (3.25)$$

We will also use the term *Nash equilibrium joint policy* for clarity. In our thesis we focus on obtaining Nash equilibria for all the agents although there exists some controversy in the literature regarding its usefulness. We can see here that the concept differs from an optimal policy. Nash equilibria represent stable points in which any individual change of policy will become a loss for the agent changing it. But it could happen that a much better individual policy exists if two agents change their policies. In any case, we have the following powerful result from game theory that helps us to search Nash equilibria:

Theorem 4 (Existence) *Given a Markov game, there exists at least one Nash equilibrium joint policy.*

In game theory terms, this theorem is usually phrased detailing that the stochastic (Markov) game can be any **general-sum** game (meaning that the sum of reward functions can have any value) and that the policies are in fact **stationary strategies** (they do not change over time; for the same state all the agents will act in the same way).

There are two main ways in which we can solve a Markov game and obtain a Nash equilibrium [31]. The first one consists on *best response* learning algorithms, in which each agent tries to learn its own best response policy against the rest of agents. In the trivial case where only one agent learns, the problem to solve becomes identical to a Markov Decision Process to all effects and therefore all the previously commented single-agent reinforcement learning algorithms would belong to this category, such as Q-learning.

Typically we deal with more complex problems in which two or more agents learn (independently) at the same time. In this case, an agent following this best-response approach may have trouble adapting its policy to what the other agents are doing. In fact, by using an algorithm suited for an MDP, the agent may be faced with the problem that the environment (from its point of view) is no longer Markov (since the transitions now not only depend on the previous state but also of all the previous ones, because the other agents have been changing their policies at each iteration), making the training more difficult.

The second main approach in solving Markov games employs *equilibrium* learning algorithms whose objective is to directly learn a Nash equilibrium for all the agents.

In our case, since we focus on approaches that always allow each agent to learn independently from the others, we have selected the *best-response* approach for problems involving two learning agents. Concretely we have chosen the MDP simplification approach, in which each agent learns by optimising its associated MDP from its own point of view. This simplification is not an uncommon approach [24], since it remarks the individuality of the learning process of each agent. Furthermore, since in our case we want a decentralised approach in which agents learn without excessively caring about the others and focus on the learning of moral values, we believe that this is at least a good starting point.

3.4 Moral values

Moral values represent ethical codes that help us discern good from wrong. Although moral values have been well studied by philosophers and sociologists [40, 50], the AI community has traditionally been less active and has not fully exploited all the ethical knowledge. The trend however has recently changed, and we can find a number of relevant contributions taking into account issues such as morality or responsibility [4, 7, 11] and even newly created conferences in Ethics and AI (like the AAAI/ACM Conference on AI, Ethics, and Society¹, which has only celebrated two editions). However, a review of the current state of the literature in Ethics and AI indicates that there are still no contributions that offer a formal definition of the concept of moral value. Our first research question tries to solve this very problem and in this section we offer the philosophical grounding that has lead us to its definition.

Moral philosophy² is the branch of philosophy that contemplates what is right and wrong. Inside of it we find the field of normative-ethics, which tries to offer an answer to *what ought we to do?*. There are three main alternative ethical theories that offer a solution to this problem: virtue ethics, utilitarianism, and deontology.

1. **Virtue ethics:** we ought to do those acts that a virtuous person would do [12].
2. **Utilitarianism:** we ought to do those acts that maximise benefit for the largest number of people [46].

¹<https://dl.acm.org/citation.cfm?id=3278721>

²<http://ethicsunwrapped.utexas.edu/glossary/moral-philosophy>

3. **Deontology**: we ought to do those acts that follow deontological principles. This theory is usually associated with philosopher Immanuel Kant [20].

Following this enunciation, there are also three main spheres in which we can evaluate the morality of committed acts:

1. The **consequentialist** one: to judge actions based on their consequences.
2. The **deontic** one: to judge actions based on duties. Part of this approach is devoted to describe the minimal conditions of morality (such as the ones that should be legally binding, like forbidding killing).
3. The **axiological** one: to judge actions based on values³. Part of this approach is devoted to describe the higher conditions of morality, which can only be recommended but not strictly required⁴ [16]. Related to that is the concept of *supererogation* [52], which we can define as going beyond the call of duty. That is, to act in ways that are morally valuable, but not necessarily required. Solidarity is a typical example of a supererogatory value.

For our formal definition of moral value we will try to combine the deontic and axiological approaches. More specifically, we take a deontic perspective to define a moral value in terms of those norms that promote it by:

- Forbidding *wrong* actions that demote it.
- Requiring *good* actions that promote it.
- Allowing *supererogatory* actions that also promote it but are not compulsory.

Of course the norms that may be enacted would depend on the situation. Furthermore, we also take an axiological perspective to expand our definition of moral value by creating an axiological function that quantifies the degree of value promotion or demotion of individual actions. With this we will give our agents a sense of what are *praiseworthy* actions when there are no regulations. We believe that this hybrid approach compromising two ethical perspectives is the most reasonable attempt to minimise their known individual weaknesses [51, 53].

³Axiology: from Greek *axia* “value, worth” and *logia* “study”: philosophical study of value.

⁴<https://plato.stanford.edu/entries/supererogation/>

Chapter 4

The Ethical Learning Problem: formalisation and theoretical results

4.1 Introduction

The objective of this chapter is to present a formal structure of a moral value (therefore answering our first research question) and to formalise our second research question (*can agents learn to behave ethically?*) as an *ethical learning problem*.

This chapter is structured as follows. Before starting with the formalisations, in section 4.2 we present a motivating example where learning ethical behaviours is needed. We call this example the *public civility problem*, being a particular case of our ethical learning problem. With this problem we can identify most of the necessary elements that our ethical learning framework must have.

After presenting this motivating example, we proceed in section 4.3 by presenting the formal definition of a moral value, which is the key piece of our entire ethical learning framework.

We continue in section 4.4 by gradually presenting our ethical learning framework (an extension of the reinforcement learning framework), first for the single-agent case by extending Markov Decision Processes (MDP) and then for the general multi-agent case by extending Markov games (MG). We continue expanding this framework with section 4.5 by analytically finding and identifying the necessary and sufficient conditions for an agent to behave ethically.

Finally, section 4.6 provides the formal definition of the ethical learning problem.

4.2 Case Study: the public civility problem

In this section we present the following motivating example: imagine a very simplified urban case study where different individuals (two for the sake of simplicity) move from their original position to their destinations as fast as possible (let's say, they are going from home to their respective jobs and they do not want to be late) through a straight lane barely wide enough to allow each of these individuals to walk on their side of the lane without disturbing each other. This pathway is usually clean so this people can walk alongside without trouble, but today an anonymous fiend has left their garbage on one side of the path instead of throwing it into the nearest wastebasket. This has hindered the usual route of one of these individuals who are going to work, and has forced them to take a decision about what should they do once they face the garbage. There are several possible actions that an individual can take at this point depending on their degree of civility, which in turn will lead to different outcomes. And of course these outcomes cannot be safely predicted since the individual is not alone in the world: perhaps by pushing the garbage to a side in order to clear their own path they may hit another passerby by doing so. We will refer to this problem as the *public civility problem*.

As an entry point, we begin by formalising the public civility problem in the simplest case: a multi-agent system formed only by two agents as figure 4.1 illustrates. The one at the left is the one faced with the problem of deciding what to do with the garbage in front, which we refer to as the **learning** agent, since it still has to learn what to do.



Figure 4.1: Left: garbage blocking the path of the agent at the left. Right: the left agent throwing the garbage. Pictures courtesy of Jordi Reyes Iso.

This learning agent is faced with the following problem: *what should I do when I encounter a garbage on my way?* Rephrased in terms of reinforcement learning, this problem is equal to searching for the optimal action in a state in which the garbage is in front. Deepening in this optimal decision making formulation, we need a model of our public civility problem that allows us to represent the following elements: the possible situations or **states** that may happen depending on the respective position of the individuals and the garbage on the street, the possible **actions** that an individual may take (not only when faced with the garbage but in any situation), the outcomes that may result after carrying out an action (which may or may not be known and which can be stochastic), or said in other words, the probability associated to each **transition** of states and actions, and finally a notion of benefit or **reward** that lets us know how good is to perform each action at each situation. Precisely reinforcement learning has a model with all of these properties: **Markov Decision Processes** (MDP).

In this example we have only mentioned one individual needing to take decisions and learn what to do, but we see that we can generalise this problem by considering that all the individuals on the street have to make decisions. In this case instead of working with a MDP, the problem can be modelled as a **Markov game** (MG). For that reason we will centre this thesis on how we can solve Markov Decision Processes and Markov games that model our public civility problem. These can slightly vary depending on the moral values we want these individuals to learn, which in turn will modify their optimal policies accordingly. Yet, MDP and MG only offer us formal models of the problem, but we still need to actually solve it, which is in fact what we ultimately aim at. There are many ways to find an optimal policy. As previously mentioned, since we focus on how an agent can learn by itself solely interacting with its environment, our choice has been to apply the rich framework of **Reinforcement Learning** (RL) algorithms.

To finish this section, we enumerate the four possible policies that the agent can follow to solve the public civility problem and the implications of each one of them.

0. **Avoidance policy.** The agent can simply dodge the obstacle, but this probably is not optimal since it takes more time by surrounding the obstacle than by pushing it to the other side. Therefore we will not consider this policy for further study.
1. **Unregulated policy.** If the agent does not respect any moral value nor cares about current norms, it can just push the garbage away whenever it encounters it to clear its path. But this may have negative effects for the other agent: dirtying, hurting it or even blocking its path (apart from also dirtying the street). Hence, this is the most blameworthy and uncivil policy.

2. **Regimented policy.** It is quite likely that hurting one another is prohibited by some norm. If a norm-compliant agent complies with this norm (even if it is only to avoid the penalty for violating it), it can learn that pushing the garbage away without considering the other agent is not a good idea at all. So it can learn that the best option is to get rid of the garbage when there is no other agent around. This is a solution a bit more civic than the first one, but we can see that there is room for improvement.
3. **Ethical policy.** An ethical agent that also complies with all possible norms can learn to do a praiseworthy action and always move the garbage to the nearest wastebasket, so it does not disturb the other agent (nor it hurts it) and also keeps the street clean. This is the most civic option.

We would like the agent to learn the last and most **ethical** policies. We will see in section 4.4 that the three latter policies serve as a basis for our ethical learning framework.

A learning agent will solve this problem by choosing the option that benefits it the most. So it will choose the unregulated policy if complying with norms and following moral values is fruitless from its perspective, the regimented policy if at least complying with norms represents a benefit (or an absence of loss), or the ethical policy if both norm and moral value compliance results in a better reward than not complying at all. Since we want the agent to learn to choose the ethical policy, in the following sections we present our novel framework that precisely tells us which are the necessary and sufficient conditions for this to happen.

4.3 Moral values formalisation

Now that we have presented the public civility problem we can begin to present our ethical learning framework. The first concept we have to introduce is of course the formal definition of a moral value. But prior to defining a moral value we first need a definition of norm. This definition is an extension of the one that we showed in the section 3.2 by including penalties, which also are very common in the MAS literature. This new definition is based on: $\mathcal{A}$, the set of actions available to each agent; $\mathcal{S}$ the set of the environment's states; and $\mathcal{L}$, a first-order logic language that allows to describe what is true and false at each state by means of expressions of the language.

Definition 16 (Norm) *A norm is a tuple $\langle c, \theta(a), p \rangle$, where c is a precondition for norm application, $\theta \in \{\text{Obl}, \text{Per}, \text{Prh}\}$ is a deontic operator regulating action $a \in \mathcal{A}$, and p is a positive value representing the punishment for non-compliance.*

Note 2 *Notice that the precondition c of a norm is a set of first-order predicates $p(\vec{\tau})$, where each p is a k_p -arity predicate symbol and $\vec{\tau} \in \mathcal{T}_1 \times \dots \times \mathcal{T}_{k_p}$ is a vector of terms, and each $\mathcal{T}_i$ is a set of terms of the first-order language $\mathcal{L}$.*

Here, in this thesis, we focus on prohibitions and obligations, so that punishment p will be applied to agents performing a prohibited action under some circumstances or to agents performing an action different from the obligatory one under other circumstances. Moreover, punishment p is considered to be a positive penalty, as for specifying the quantity that will be discounted from the agent's reward function upon non-compliance.

With norms defined, we can now go one step forward and define moral values. Moral values represent ethical codes that help us discern good from wrong as explained in section 3.4. Based in the main spheres of moral philosophy, in our definition of moral value a constituent part of it are norms that promote it (based on the deontic sphere). Since moral values also provide ethical criteria to assess if a non-regimented action is actually good or wrong (according to the axiological sphere), the second constituent part of a moral value is an evaluation function that signals how good (praiseworthy) are actions according to the moral value. Formally, we define a moral value as:

Definition 17 (Moral value) A moral value v is a pair $\langle \mathcal{N}_v, f_v \rangle$ such that:

- $\mathcal{N}_v$ is a finite set of norms.
- f_v is an action evaluation function that, for an action ‘ a ’ and a precondition c (expressed in a first-order language $\mathcal{L}$), returns a number in the interval $[-1, 1]$ meaning the degree of promotion/demotion of that action to the moral value ($f_v(a, c) = 1$, means that ‘ a ’ strongly promotes the moral value given the precondition c , whereas $f_v(a, c) = -1$ means ‘ a ’ strongly demotes it).

where $\mathcal{N}_v$ and f_v satisfy the following consistency constraint:

- If there exists an action ‘ a ’ that promotes the moral value v , then there cannot be a norm in that value prohibiting the very same action. Formally, $\forall a$ s.t. $f_v(a, c) > 0$ then, $\nexists n \in \mathcal{N}_v$ s.t. $n = \langle c, \text{Prh}(a), p \rangle$.

Therefore, moral values act as inner mechanisms that help an agent to take individual decisions when faced with regulated and also unregulated situations.

4.4 Ethical learning frameworks

The objective of this section is to present a framework that extends the reward functions of agents so they are aligned with moral values and also to show how the framework should be used.

Section 4.4.1 introduces basic definitions that will become handy in the next subsections to define the necessary concepts. In subsection 4.4.2 we present the ethical learning framework for a single learning agent, which essentially consists on extending Markov Decision Processes into what we call Ethical Markov Decision Process (a subclass of them in which the reward function is aligned with a moral value). In subsection 4.4.3 we present the correspondent ethical learning framework for the multi-agent case. Similarly, it consists on extending Markov games into what we call Ethical Markov games. Finally, in subsection 4.4.4 we present theoretical results of which are the necessary and sufficient conditions for an agent to learn an ethically-aligned policy when using our ethical learning framework.

4.4.1 Basic definitions

The objective of the auxiliary functions shown here is to permit the use in a Markov Decision Process or a Markov game of norms and moral values as we have defined them. In this section we present the condition function, that allows us to go from the world of states to the world of predicates, and the penalty function that allows us to use norms in Markov games.

Definition 18 (Condition function) Given a Markov game with a set of states $\mathcal{S}$ and a first-order language $\mathcal{L}$, with its associated set of predicates $\mathcal{P}(\mathcal{L})$, we define the Condition function $C : \mathcal{S} \rightarrow 2^{\mathcal{P}(\mathcal{L})}$ that maps every state to the set of predicates of describing the state.

Moreover, whenever all the predicates in the precondition of a norm hold, then the deontic part of the norm must hold.

Definition 19 (Penalty function) Given a norm $n = \langle c, \theta(x), p \rangle$, and a Markov game with a set of states $\mathcal{S}$ and a set of actions $\mathcal{A}$, we define the penalty function $P_n : \mathcal{S} \times \mathcal{A} \rightarrow \{0, p\}$ as

$$P_n(s, a) \doteq \begin{cases} p & \text{if } c \in C(s), \theta = \text{Prh} \text{ and } a = x \\ & \text{or if } c \in C(s), \theta = \text{Obl} \text{ and } a \neq x, \\ 0 & \text{otherwise.} \end{cases} \quad (4.1)$$

We call this the penalty function because it tells us in which situations, if a given norm is in force, an agent would receive a penalty for violating it and what is the value of such penalty.

4.4.2 Single-agent Ethical Learning Framework

Before defining the main concept of this section, the Ethical Markov Decision Process, we first present an intermediate concept, the Normative Markov Decision Process, in which the reward function penalises the violation of norms. This later serves as a basis for our Ethical Markov Decision Process definition, since we have defined moral values as two components, one of them being a set of norms.

Definition 20 (Normative Markov Decision Process) *Given a set of norms $\mathcal{N}$ and a Markov Decision Process with a reward function R_0 , we define its normative extension, the so-called Normative Markov Decision Process (NMDP), as a MDP whose reward function R is obtained as $R = R_0 + R_{\mathcal{N}}$, where $R_{\mathcal{N}} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^-$ is a function such that*

$$R_{\mathcal{N}}(s, a) \doteq - \sum_{n \in \mathcal{N}} P_n(s, a). \quad (4.2)$$

We will refer to $R_{\mathcal{N}}$ as the normative function of the set of norms $\mathcal{N}$, which accumulates the penalties of all the norms enforced in a given state-action pair $\langle s, a \rangle$.

A Normative Markov Decision Process is a subclass of MDP's whose reward function penalises the violation of norms. This still does not tell us much about the kind of processes it can model. It might be the case that a NMDP does not have any policy without violating any norm, which would be beyond the scope of this work. For that reason we define an even more specific kind of MDP:

Definition 21 (Feasible Normative MDP) *We say that a Normative MDP of a set of norms $\mathcal{N}$ is feasible if and only if for every state $s \in \mathcal{S}$ there exists at least one action $a \in \mathcal{A}$ such that $R_{\mathcal{N}}(s, a) = 0$.*

Note 3 *The definition of Feasible Normative MDP is equivalent to saying that, for any state of the MDP, there is always an action available to the agent that complies with all the norms in force. This is a minimality condition.*

Now we can expand this model to include moral values.

Definition 22 (Ethical Markov Decision Process) *Given a moral value $v = \langle \mathcal{N}_v, f_v \rangle$ and a Markov Decision Process with a reward function R_0 , we define its ethical extension, the so-called Ethical Markov Decision Process (EMDP), as an MDP whose reward function R is defined as $R = R_0 + R_{\mathcal{N}_v} + R_v$, where $R_{\mathcal{N}_v} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^-$ is the normative function associated to norm set $\mathcal{N}_v$ and $R_v : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^+$ is a function of the form*

$$R_v(s, a) \doteq \mu(s) \cdot \max(0, f_v(a, C(s))), \quad (4.3)$$

where $\mu : \mathcal{S} \rightarrow (0, \infty)$ is a strictly positive function. We will refer to R_v as the ethical function of moral value v , which rewards praiseworthy actions.

Note 4 *Notice that R_v is the adaptation of f_v from the world of first-order languages to the world of states. The μ is a function dependant of the MDP as a weight so that the value of f_v (which is bounded in $[0, 1]$) can be significant compared with the rest of rewards of the MDP. An example: it is useless to reward with 1 point a praiseworthy action if blameworthy actions give a reward of 1000.*

Note 5 *Notice that an Ethical MDP is trivially a Normative MDP by rearranging the reward function as $(R_0 + R_v) + R_{\mathcal{N}_v}$.*

The definition of an Ethical MDP is general enough to allow Ethical MDP's where an agent can never perform ethically-aligned actions. For that we define the concept of an Attainable Ethical MDP.

Definition 23 (Attainable Ethical MDP) *Given an Ethical MDP for moral value $v = \langle \mathcal{N}_v, f_v \rangle$, we say that it is attainable if the NMDP with the reward function $(R_0 + R_v) + R_{\mathcal{N}_v}$ is feasible and if there exists at least one tuple $\langle s, a \rangle$ such that $R_v(s, a) > 0$.*

4.4.3 Multi-agent Ethical Learning Framework

Now we will expand the definitions we have presented to be also available for use in Markov Games. We need to define the notions of the counterparts of NMDP and EMDP in a multi-agent context.

Definition 24 (Normative Markov game) *Given a set of norms $\mathcal{N}$ and a Markov game of m agents with reward functions $R_0^1, \dots, R_0^m$, we define its normative extension, the so-called Normative Markov game (NMG), as a Markov game such that each reward function R^i is defined as $R^i = R_0^i + R_{\mathcal{N}}^i$, where $R_{\mathcal{N}}^i : \mathcal{S} \times \mathcal{A}^i \rightarrow \mathbb{R}^-$ is a normative function of norm set $\mathcal{N}$.*

Definition 25 (Feasible Normative Markov game) *We say that a Normative Markov game of m agents is feasible if and only if for every state s there exists at least one joint action $\langle a^1, \dots, a^m \rangle$ such that $R_{\mathcal{N}}^i(s, a^i) = 0$ for every i .*

And now we proceed with Ethical Markov games, likewise the previous subsection.

Definition 26 (Ethical Markov game) *Given a moral value $v = \langle \mathcal{N}_v, f_v \rangle$ and a Markov game of m agents with reward functions $R_0^1, \dots, R_0^m$, we define its ethical extension, the so-called Ethical Markov game (EMG), as a Markov game such that each reward function R^i is defined as $R^i = R_0^i + R_{\mathcal{N}_v}^i + R_v^i$, where $R_{\mathcal{N}_v}^i : \mathcal{S} \times \mathcal{A}^i \rightarrow \mathbb{R}^-$ is the normative function of norm set $\mathcal{N}$ applied to agent i and $R_v^i : \mathcal{S} \times \mathcal{A}^i \rightarrow \mathbb{R}^+$ is the ethical function of moral value v applied to agent i .*

Definition 27 (Attainable Ethical Markov game) *Given an Ethical Markov game of a moral value $v = \langle \mathcal{N}_v, f_v \rangle$ and m agents, we say that it is attainable if the NMG with reward functions $(R_0^i + R_{\mathcal{N}_v}^i) + R_v^i$ is feasible and if there exists at least one tuple $\langle s, \{a^1, \dots, a^m\} \rangle$ such that $R_v^i(s, a^i) > 0$ for every i .*

Now that we have defined what an Attainable Ethical Markov game is, we would like to have a measure to assess how ethically aligned is a policy learnt by an agent towards a moral value. The next subsection provides our answer to that.

4.4.4 Policy evaluation

The objective of this subsection is to formalise what it means for a policy to be ethically-aligned. For that we first need to define some new concepts such as the Normative fulfilment degree and the Ethical fulfilment degree. Without loss of generality we will work in this section always with Markov games.

Our first definition is the Normative fulfilment degree of a policy. Informally, it represents how much an agent follows regulations.

Definition 28 (Normative fulfilment degree) *Given a Markov game with m agents, a finite set of norms $\mathcal{N}$ and a policy π , we define the Normative fulfilment degree $NF(\pi, \mathcal{N}) \in [0, 1]$ of policy π as*

$$NF(\pi, \mathcal{N}) \doteq \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \sum_{a \in \mathcal{A}} \pi(s, a) \cdot (1 + \text{sgn}(R_{\mathcal{N}}(s, a))), \quad (4.4)$$

where $\mathcal{S}' \subset \mathcal{S}$ is defined as

$$\mathcal{S}' \doteq \{s \in \mathcal{S} \mid \min_{a' \in \mathcal{A}} R_{\mathcal{N}}(s, a') < 0\}, \quad (4.5)$$

and $\text{sgn} : \mathbb{R} \rightarrow \{-1, 0, 1\}$ is the sign function that assigns -1 to any negative number, 1 to any positive number and 0 to itself.

This degree averages for how many states (of the subset of states $\mathcal{S}'$ in which norms are violable), the policy complies with all the norms in force. A value $NF = 1$ implies that the agent is always following all the norms, whereas a value $NF = 0$ implies the complete opposite. We can simplify the measure in case of the policy being deterministic.

Example 1 *If a policy π is deterministic, then the policy followed by an agent becomes $\pi'(s) \doteq \arg \max_a \pi(s, a)$, and the previous formula gets simplified into:*

$$NF(\pi', \mathcal{N}) \doteq \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} (1 + \text{sgn}(R_{\mathcal{N}}(s, \pi'(s)))), \quad (4.6)$$

where $\mathcal{S}'$ is as in equation 4.5.

This measure allows us to enunciate this interesting definition for checking policies that are norm-compliant.

Definition 29 (Norm-compliant joint policy) *Given a Normative Markov game with m agents, and a set of norms $\mathcal{N}$, we say that $\prod_i \pi^i$ is a norm-compliant joint policy if $\prod_{i=1}^m NF(\pi^i, \mathcal{N}) = 1$.*

And now we can introduce the equivalent measure for moral values, the Ethical fulfilment degree:

Definition 30 (Ethical fulfilment degree) *Given a Markov game with m agents, a moral value v and a policy π , we define the Ethical fulfilment degree $EF(\pi, v) \in [0, 1]$ of policy π as*

$$EF(\pi, v) \doteq \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \sum_{a \in \mathcal{A}} \pi(s, a) \frac{R_v(s, a)}{\max_{a' \in \mathcal{A}} R_v(s, a')}, \quad (4.7)$$

where $\mathcal{S}' \subset \mathcal{S}$ is defined as

$$\mathcal{S}' \doteq \{s \in \mathcal{S} \mid \max_{a' \in \mathcal{A}} R_v(s, a') > 0\}. \quad (4.8)$$

The Ethical fulfilment degrees averages for how many states (of the subset of states $\mathcal{S}'$ in which praiseworthy actions are available), the policy recommends doing a praiseworthy action and how good it is. Therefore, similar to the normative fulfilment degree, $EF = 1$ implies that for a policy π , the policy is totally aligned with the moral value v and supports it every time there is a chance for it, whereas a value $EF = 0$ implies that the agent never supports the moral value. Notice that this measure says nothing about the times when the agent acts *against* the moral value.

Example 2 *If a policy is deterministic, then we can express $\pi'(s) \doteq \arg \max_a \pi(s, a)$ and the previous formula gets simplified into:*

$$EF(\pi', v) \doteq \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \frac{R_v(s, \pi'(s))}{\max_{a'} R_v(s, a')}, \quad (4.9)$$

where $\mathcal{S}'$ is as in equation 4.8.

This definition also allows us to define the notion of ethically-consistent policies.

Definition 31 (Ethically-consistent policy) *Given a Markov game and a moral value v we say that a joint policy $\prod_i \pi^i$ is ethically-consistent if $\prod_i NF(\pi^i, \mathcal{N}_v) = 1$ and $\prod_i EF(\pi^i, v) > 0$.*

Note 6 Given a Markov game and a moral value v , a joint policy $\pi = \prod_i \pi^i$ such that

$$\prod_i NF(\pi^i, \mathcal{N}_v) \cdot EF(\pi^i, v) = 1, \quad (4.10)$$

is an ethically-consistent policy that also is totally aligned with the moral value v .

This note tells us indirectly how to set the normative and ethical functions of an Ethical Markov game so the optimal policy is also totally aligned with the moral value. We should define the penalties p and ethical reward factors μ of an Ethical Markov game in such a way that the policies $\pi_\star^i$ that the agents learn are also what we define as ethically-aligned policies.

Definition 32 (Ethically-aligned policy) Given a Markov game and a moral value v we define a policy π such that

$$NF(\pi, \mathcal{N}_v) \cdot EF(\pi, v) = 1, \quad (4.11)$$

as an ethically-aligned policy.

We will also use the term *ethically-aligned joint policy* when every agent follows an ethically-aligned policy. We are now prepared to enunciate and prove in the next section the theorem of characterisation of ethically-aligned policies.

4.5 Characterising ethically-aligned policies

The objective of this section is to prove subsequent theorems 5 and 6, which formalise the sufficient and necessary conditions for an agent to learn to behave ethically-aligned with a moral value in a Markov Decision Process and a Markov game, respectively. These theorems also characterise the policies that the agents need to learn, which we can assume to be deterministic.

Lemma 1 (Norm-compliant policy characterisation) Given a Markov Decision Process and a set of norms $\mathcal{N}$, then a deterministic policy $\pi_\star$ is also norm-compliant if and only if for every state s it holds that $a_\star = \arg \max_a \pi(s, a)$ is such that

$$R_{\mathcal{N}}(s, a_\star) = 0. \quad (4.12)$$

Proof 1 On the one hand, $R_{\mathcal{N}}(s, a_\star) = 0$ for every state s directly implies that $NF(\pi_\star, \mathcal{N}) = 1$, and therefore that $\pi_\star$ is a norm-compliant policy. On the other hand we prove it by *reductio ad absurdum*. Suppose that for the optimal policy $\pi_\star$ there exists a state s such that $a_\star = \arg \max_a \pi(s, a)$ and also $R_{\mathcal{N}}(s, a_\star) < 0$. This would mean that $\pi_\star$ has a $NF(\pi_\star, \mathcal{N}) < 1$, which is a contradiction with being norm-compliant.

With this we can determine when an optimal and deterministic policy is also ethically-aligned in a Markov Decision Process.

Theorem 5 (Ethically-aligned policy characterisation) Given a Markov Decision Process and a moral value v , then a deterministic optimal policy $\pi_\star$ is also ethically-aligned if and only if for every state s it holds that $a_\star = \arg \max_a \pi(s, a)$ is such that

$$(i) \ a_\star = \arg \max_a R_v(s, a) \quad \text{and}$$

$$(ii) \ R_{\mathcal{N}_v}(s, a_\star) = 0.$$

Proof 2 On the one hand, $a_* = \arg \max_a R_v(s, a)$ implies that $EF(\pi_*, v) = 1$ and $R_N(s, a_*) = 0$ implies that $NF(\pi_*, \mathcal{N}) = 1$, and therefore that π_* is a ethically-aligned policy. On the other hand we prove it again by reductio ad absurdum. Suppose that for the optimal policy π_* there is a state s such that $a_* = \arg \max_a \pi(s, a) \neq \arg \max_a R_v(s, a)$, then there exists an action $b \neq a_*$ such that $R_v(s, a_*) < R_v(s, b)$, which would imply that $EF(\pi_*, v) < 1$, which is a contradiction with being ethically-aligned. The second statement is already proven by lemma 1.

Now we expand these results for the Markov game settings.

Lemma 2 (Norm-compliant joint policy characterisation) *Given a Markov game of m agents and a set of norms $\mathcal{N}$, then a deterministic joint policy π_* is also norm-compliant if and only if for every state s and every agent i it holds that $a_*^i = \arg \max_{a^i} \pi_*^i(s, a^i)$ is such that*

$$R_{\mathcal{N}}^i(s, a_*^i) = 0. \quad (4.13)$$

Proof 3 The lemma is proven following the same steps as in the proof for lemma 1.

Theorem 6 (Ethically-aligned joint policy characterisation) *Given a Markov game of m agents and a moral value v , then a deterministic joint policy π_* is also ethically-aligned if and only if for every state s and every agent i it holds that $a_*^i = \arg \max_{a^i} \pi_*^i(s, a^i)$ is such that*

- (i) $a_*^i = \arg \max_a R_v^i(s, a)$ and
- (ii) $R_{\mathcal{N}_v}^i(s, a_*^i) = 0$.

Proof 4 The lemma is proven following the same steps as in the proof for Theorem 5.

The conclusion from the final theorems 5 and 6 is that in order to ensure that the optimal or Nash equilibrium policy of an agent (that is, the policy that it is going to learn) is also ethically-aligned, we need to create penalties that make violating them a worst long-term decision than complying with them, and also to set the μ component of the ethical reward function so that being ethical compensates the long-term losses of acting ethically. This compensation is done through extending the reward function of the MDP or MG into an Ethical MDP or MG, in which the rewards for the ethical and normative functions (see definitions 20 and 22) are given by the theorem 5 and 6.

4.6 Research problems

As we have mentioned, in this master thesis we focus on aligning the policies of the agents with a given moral value. Our first research question consisted in formalising moral values, which we have already done in section 4.3. Our second research question was the problem of finding an ethically-aligned policy given that moral value formalisation. Now that we have introduced the concepts of Ethical MDP and Ethical MG, and we have formally defined an ethically-aligned policy, we can finally formalise our second research question as two research problems: one for the single-agent case and another one for the multi-agent case.

Definition 33 (Single-agent Ethical Learning Problem) *Let v be a moral value and a Markov Decision Process for an agent. The Single-agent Ethical Learning Problem is defined as learning an ethically-aligned policy.*

Note 7 *This implies that given a MDP, we are asking the agent to solve it while also complying with all the current regulations and always choosing the most ethical option (i.e., aligned to a given moral value) when faced with a moral decision. These three tasks are strongly related with the three reward functions that form an Ethical MDP: the baseline one, the normative one, and the ethical one.*

We can extend and reformulate this problem for the case with more than one learning agent:

Definition 34 (Multi-agent Ethical Learning Problem) *Let v be a moral value and M a Markov game. The Multi-agent Ethical Learning Problem is defined as learning an ethically-aligned joint policy for all the agents independently.*

Note 8 *Bear in mind that a Nash equilibrium joint policy can be obtained even if all the agents are learning independently from one another and similarly the term joint policy does not imply in any way that the agents cooperate between them (nor that they have opposing individual objectives). Hence, this problem is simply a reformulation of the previous one for every agent independently.*

In general we will speak about the *ethical learning problem* when referring to the multi-agent ethical learning problem (even for the case $m = 1$).

In this project we will solve these two problems for the case study we have presented in section 4.2, the public civility problem. In the next chapter we formalise this as a Markov game and explain all the technical details about it.

Chapter 5

Formalising the public civility problem

5.1 Introduction

In this chapter we formalise the public civility problem applying the ethical learning framework defined in Chapter 4 so that we can evaluate it in subsequent chapter 6.

This chapter is organised as follows. In section 5.2 we present the environment that models the public civility problem. Subsequently, Section 5.3 presents how we can model this environment as a Markov game by specifying the possible states, actions and rewards.

5.2 Simulation of the public civility problem

In the aforementioned public civility problem (see Sect. 4.2), two individuals move from their original positions towards their destinations with the aim of reaching their goal positions as fast as possible while also having to deal with garbage obstacles. As its name suggests, the moral value of this example is **civility**.

Figure 5.1 depicts how we model this example as a multi-agent system consisting on a 2-dimensional grid, where two agents traverse grey cells in their way towards their destination. For illustrative purposes, we represent agents as black circles –labelled as L (Left) and R (Right)– whose starting positions are the ones depicted in the figure and their destination (Goal) cells appear marked as GL and GR respectively. Moreover, two agents cannot populate the same cell simultaneously (the first one occupying a cell prevents the other’s movement). Initially, the garbage –which is depicted as a purple square– is randomly located at any of the grey cells except for the initial positions of the agents.

Agents can deal with garbage in the following ways:

- By detouring it (although this will not change the blockade nor the dirtiness and the problem will remain).
- By throwing the garbage to any other grey cell (which will change the blockade position and may even hurt the other agent if unfortunately located there).
- By pushing the garbage further to the red cells so that it does not block the way (although the street would still be dirty).
- By pushing the garbage to any of the wastebaskets on the green cells (which is the ethical thing to do because it removes the blockade and cleans the street).

Obviously, those behaviours leading to a clean street where no-one is injured will be aligned with our *civility* moral value. However, moving the garbage means to perform an effort, and thus the agents face the problem of choosing between a policy that just minimises their trip time or another policy –that will be the expected one from an exemplary citizen– that still tries to reach the individual goal as effectively as possible, but it is also aligned with the moral value of *civility*.

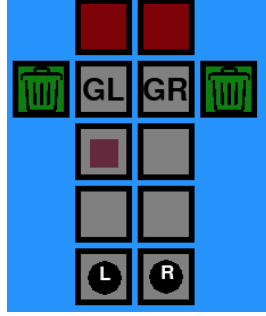


Figure 5.1: Initial screen-shot of the simulation with a randomly located garbage (purple square).

Time is discrete in our multi-agent system and measured in time-ticks for simplicity. A day (which lasts for Max_t ticks at most) corresponds to the period of time both agents need to reach their destinations. Every tick agents are allowed perform a single action: moving or pushing the garbage if it is located in front. Both movement and pushing actions can be done to adjacent cells and in three different directions: Left, Front, and Right. We have not considered backward actions for simplicity.

5.2.1 Scenarios in the public civility problems

In order to study the impact of norms and moral values in our public civility problem and put in practise the ethical framework we have developed, we have defined four different scenarios of increasing complexity. This way we will test firsthand the difference between considering norms, moral values or having no regulations in a learning multi-agent system. The scenarios are:

1. *Unregulated*: A (wild) baseline scenario with neither norms nor moral values. In this scenario, the learning problem of the agents is modelled as a basic Markov game without considering norms nor moral values.
2. *Regimented*: A scenario regulated by a norm n_1 that prohibits to throw garbage to another agent (that is, to push it towards a neighbouring cell where this agent is located) to avoid aggressive behaviours so no agent gets hurt. Formally, n_1 is defined as

$$n_1 = \langle (\text{adjacent_agent}, \text{garbage_in_front}), \text{Prh}(\text{throw_to_agent}), \mathbf{p}_1 \rangle, \quad (5.1)$$

following the structure of a norm defined in def. 16. In this scenario, the learning problem of the agents is therefore modelled as a Normative Markov game with $\mathcal{N} = \{n_1\}$.

3. *Ethical*: A scenario that is more sophisticated than the previous ones because it considers the *civility* moral value v_{civ} that: promotes the action of throwing the garbage to the wastebasket and considers that throwing it to other agents is inadmissible. Thus, we can include norm n_1 in Eq. 5.1 into N_v so to formalise civility following definition 17 as:

$$v_{civ} = \langle \{n_1\}, f_{civ} \rangle, \text{ where: } f_{civ}(\text{put_garbage_in_wastebasket}, \text{garbage_in_front}) = \text{eval}_{civ}, \quad (5.2)$$

and $f_{civ}(a, c) = 0$ otherwise. Remember that f_{civ} is the evaluation function of the moral value and $\mathbf{eval}_{civ} > 0$ is a positive constant. In this scenario, the learning problem of the agents is consequently modelled as an Ethical Markov game with $v = v_{civ}$.

4. *Over-regimented:* A (possibly over-regulated) scenario that changes the recommendations of value-aligned actions that promote the moral value so to enact them as norms. In this scenario, agents are regulated by the previously defined n_1 from equation 5.1 and a new norm n_2 that obligates them to throw the garbage to the wastebasket when found. Agents that do not comply with norm n_2 will receive a punishment p_2 . Formally, n_2 is defined as:

$$n_2 = \langle (\text{garbage_in_front}), \text{Obl}(\text{throw_to_wastebasket}), p_2 \rangle, \quad (5.3)$$

following the structure of a norm defined in def. 16. In this scenario, the learning problem of the agents is therefore modelled as a Normative Markov game, like the regimented one, but with $\mathcal{N}' = \{n_1, n_2\}$.

The unregulated scenario serves as a baseline for comparison with all the rest. Theoretically, the regimented scenario should lead to more civic behaviours than the unregulated one, and in the ethical scenario agents should behave even more civilly. We have added a fourth over-regimented scenario very similar to the ethical one but with the difference of only having norms and no moral values.

5.3 An Ethical Markov game for the public civility problem

Since we aim to solve the Markov game modelling for the *Public Civility Problem* by applying reinforcement learning algorithms, we need to model our problem as a Markov game. In this subsection we specify the states, actions, transition function and reward functions in each scenario accordingly.

5.3.1 States

We define a state $s \in \mathcal{S}$ as a tuple

$$s = \langle cell_L, cell_R, cell_G \rangle, \quad (5.4)$$

where $cell_L$ corresponds to the position (cell) of agent L; $cell_R$ is the position of agent R; and $cell_G$ identifies the position of the garbage.

Considering that there are 8 possible passable cells for the agents (but they cannot be at the same one) whereas the garbage has a total of 10 possible cell positions (it can be at all the passable cells except the agents' initial ones plus the four coloured ones), there is a total of $8 \cdot 7 \cdot 10 = 560$ possible states.

Terminal states exist and are defined by the discrete function $f_F : \mathcal{S} \times \mathbb{N} \rightarrow \{0, 1\}$. In our case, terminal states include those where both agents reach their goal positions (regardless of the position of the garbage) as well as states occurring when the current simulation reaches the maximum number of ticks (Max_t):

$$f_F(s, t) \doteq \begin{cases} 1 & \text{if } (s.cell_L = GL \ \& \ s.cell_R = GR) \text{ or } t \geq Max_t, \\ 0 & \text{otherwise.} \end{cases} \quad (5.5)$$

where t corresponds to the current simulation tick and $s.cell_L$ and $s.cell_R$ respectively represent the cell positions of agent L and R at state s . Max_t is related to the number of ticks days one day lasts (being Max_t the maximum number of ticks per day).

5.3.2 Actions

As aforementioned, the set of actions each agent can perform in every scenario is

$$\mathcal{A} = \{mF, mR, mL, pF, pR, pL\}, \quad (5.6)$$

where *m* stands for **m**ovement and *p* for **p**ush. Moreover *F*=Forward, *R*=Right, and *L*=Left. Actions *mF*, *mR*, and *mL* imply a change in the agent position ($s.cell_L$) (if the action makes sense; obviously moving to the left will not change its position if there is no cell at the left), whereas actions *pF*, *pR*, and *pL* will change the garbage's position ($s.cell_G$) whenever the garbage is in front of the agent.

Any action $a \in \mathcal{A}$ can be performed at any state $s \in \mathcal{S}$, meaning that there are 6 possible actions for each state, adding up to a total of 3360 action-state pairs. This number includes all the dummy combinations with no effect on the environment. Next we detail when each action has no effect.

- *mF*: if there is an obstacle in front of the agent. Obstacle refers both to the garbage and to another agent. Also it has no effect if there is no passable empty grey cell in front.
- *mR*: if there is an obstacle in the contiguous right cell or there is no passable grey cell on the right.
- *mL*: if there is an obstacle in the contiguous left cell or there is no passable grey cell.
- *pF*: if there is no garbage to push or if there is no cell to push it in front.
- *pR*: if there is no garbage to push or if there is no cell to the right to push it.
- *pL*: if there is no garbage to push or if there is no cell to the left to push it.

There is only one more case of dummy action not listed above: if two agents try to go to the same (unoccupied) cell at the same time, only one of them will get there while the other will stand still. The agent to move is chosen randomly.

5.3.3 Transition function

Transitions are fully deterministic in our public civility problem. This means that $T(s, a^L, a^R, s') \in \{0, 1\}$ for every possible value, where $a^L \in \mathcal{A}^L$ is L agent's action and $a^R \in \mathcal{A}^R$ is R agent's action. This can be misleading, because whereas it is true that transitions are fully deterministic, agents act independently from each other meaning that they cannot know for certain which will be the next state when executing their action.

So from the point of view of each agent, transitions are actually non-deterministic. For instance, from the L agent's point of view, transitions follow the formula

$$T(s, a^L, s') = \sum_{a^R \in \mathcal{A}} \pi^R(s, a^R) \cdot T(s, a^L, a^R, s'), \quad (5.7)$$

which can have positive values smaller than 1. The same happens from the R agent's point of view.

5.3.4 Reward Functions

Since state transitions are deterministic, so are rewards given. Considering $s \in S$ to be the current state, $a^L \in \mathcal{A}$ the action agent L performs, a^R the action agent R performs and $s' \in \mathcal{S}$ such that $\langle s, a^L, a^R, s' \rangle$ is a transition, we define a different deterministic reward function $R^i(s, a^L, a^R, s')$ for each scenario and agent, with $i \in \{L, R\}$ to identify the agent that it is associated with. We do so by extending the reward function in a way that shapes agents' policies with ethical knowledge $R^i = R_0^i + R_{\mathcal{N}_v}^i + R_v^i$ following definitions 20 and 22.

First, in the *Unregulated* scenario each agent i just minds reaching its respective destination Gi (GR or GL) as fast as possible, thus

$$R_0^i(s, a^L, a^R, s') \doteq \begin{cases} Max_t & \text{if } s'.cell_i = Gi, \\ -1 & \text{otherwise.} \end{cases} \quad (5.8)$$

This formulation is done to encourage that each agent reaches its goal position as soon as possible, since by penalising the agent for being in any position except its goal, we are encouraging it to never stop until it gets to its goal. It is important to remark that other formulations may be perfectly valid as well.

Second, the *Regimented* scenario is regulated by norm n_1 in Eq. 5.1, and thus, the agents will be punished with p_1 if they do not comply with n_1 . Thus, we define the following normative function instantiating Eq. 4.2:

$$R_{\mathcal{N}}^i(s, a^i) = -P_{n_1}^i(s, a^i). \quad (5.9)$$

Agents in this scenario receive a total reward that corresponds to the addition of both reward functions R_0^i and $R_{\mathcal{N}}^i$.

Third, for the *Ethical* scenario we define an *Ethical function* R_v^i (see Eq. 4.3) for the *civility* moral value (v_{civ} in Eq. 5.2) that rewards those that behave civilly by keeping the streets clear (and clean). In the Ethical scenario there is also a *Normative function* $R_{\mathcal{N}_v}^i$ identical to the previously defined one, that avoids violence by punishing violators of n_1 . Specifically, the ethical function is defined as:

$$R_v^i(s, a^i) = \mu(s) \cdot \max(0, f_v(a^i, C(s))). \quad (5.10)$$

where $f_v(a^i, C(s))$ only returns $eval_{civ}$ from eq. 5.2 if agent i performs any garbage pushing action pF , pR or pL that will move the garbage to any of the wastebaskets, and returns 0 otherwise. As for our previous scenario, the agents will receive an extended reward of $R_0^i + R_{\mathcal{N}_v}^i + R_v^i$.

Finally, the *Over-regimented* scenario is regulated by $\mathcal{N}' = \{n_1, n_2\}$. Thus, this time the normative function is:

$$R_{\mathcal{N}'}^i(s, a^i) = -(P_{n_1}^i(s, a^i) + P_{n_2}^i(s, a^i)). \quad (5.11)$$

Agents in this scenario receive a total reward that corresponds to the addition of both reward functions R_0^i and $R_{\mathcal{N}'}^i$.

Chapter 6

Evaluation

6.1 Introduction

This chapter is devoted to evaluate our ethical learning framework in the public civility problem. In section 6.2 we explain which were our hypotheses prior to the experiments. Then, in section 6.3 we present the methodology we have followed to evaluate our framework. Finally, in section 6.4 we show which are the results of the evaluation.

6.2 Goals (hypotheses)

We have two differentiated goals or empirical hypotheses. Essentially they can be summarised in this sentence: *the agents will (1) achieve their individual objectives as fast as they can while also (2) obtaining an ethically-aligned policy if they solve the Ethical Markov game*. Let us explicitly detail them:

- The first and more classical one is the assessment of the obtention of an optimal policy that minimises the time the agents need to achieve their individual objectives, related with our reinforcement learning evaluation.

This is going to be evaluated via displaying the accumulated reward the agents obtain while training and also via analysing their learnt policies with two measures: the average time taken to fulfil their objectives and their level of tiredness when doing so. They will be explained in detail in the subsection 6.3.2.

- The second goal is strongly related with the third research problem of our master thesis (*Can we assess the social benefits derived from the ethically-aligned behaviours of agents?*). We would like to observe that as the agents learn a more ethical policy (a civic policy in our case), the society they constitute improves as a whole.

This improvement will be measured both from an individual perspective (of each agent) and from a global perspective (of the society).

6.3 Methodology

This section is devoted to explain the methods we use to test our hypotheses. In subsection 6.3.1 we make an explicit list of all the learning settings of the public civility problem that we consider for evaluation. In subsection 6.3.2 we present all the measures that we apply to evaluate the policies that the agents learn at each setting. In subsection 6.3.3 we detail which algorithm the agents use in order

to learn their policies. Finally, in subsection 6.3.4 we enumerate all the tests that we will perform to the policies that the agents learn.

6.3.1 Simulation learning settings

In our public civility problem, if an agent pushes the garbage towards the lane of the other agent, it is solving its problem today, but tomorrow when the other agent pushes the garbage back to its lane the problem will reappear, continuing with the problem for another day. This is known as a sequential social dilemma (that has recently started to be studied in multi-agent reinforcement learning [24, 35]). As a test to see how this *continuity* affects the learning of the agents policies, we have considered two studies of the public civility problem:

- *Episodic*: where the agents' actions taken one day (episode) do not affect the next one. Therefore, all the episodes are independent.
- *Sequential*: the agents' actions taken one day (episode) affect the next one in the sequence, so days are not independent. Each sequence lasts for $d \gg 1$ days (episodes).

The only difference between both is that in the sequential version, the garbage remains at the same position the next day (if the previous one was not the last day of the episode). We consider these two alternative periods in order to analyse if long-term effects (such as leaving the garbage on the street) influence the learnt policy of the agents on each scenario. Thus, in total, we study two variations for each of the four scenarios defined in subsection 5.2.1.

Each of these eight scenario cases is tackled by default with the two agents applying reinforcement learning to learn which is their respective optimal policy to deal with the problem. But we can also consider that the number of learning agents is actually another parameter modifying the environment that can have two values:

1. A single learning agent environment where just the left agent L learns through reinforcement learning while the right R agent uses a pre-programmed policy that we detail in section 6.3.3. This approach allows us to study how a newcomer adapts to an already organised society and also serves as a control version for the next one.
2. A multi-agent learning environment where both agents apply reinforcement learning. This approach allows us to study the emergence of social consensuses.

In practise, we will evaluate 16 different settings, each of them being individually studied with the measures that we present in the following subsection.

Simulation parameters setup

When considering the eight different scenario studies, several parameters need to be set up. In our experiments, we consider the following parameter values for the environment model:

- The amount of consecutive episodes (days) d in the sequential scenario variations will be set to $d = 10$ for simplicity.
- The maximum amount of time-ticks per episode will be $Max_t = 20$. Considering that an agent would need 6 time-clicks at max if faced with the garbage in front of it at the beginning of an episode and wanted to push it to garbage, we believe this is a reasonable amount of time. Accordingly, the positive reward in the baseline unregulated reward function in Eq. 5.8 will also be of $Max_t = 20$.

And we consider the following parameters for the reward functions:

- In the regimented scenario reward function, we set a punishment of $p_1 = 10$ for not complying with norm n_1 (see equation 5.1). An agent compliant with this norm may lose up to 3 time-ticks in the worse scenarios, so if the norm violation results in a relative reward decrease of $|3 - 10| = |-7|$, it will dissuade the agent from violating the norm
- In the ethical scenario reward function we set a reward of 10 for behaving civilly computed as the product of $\text{eval}_{\text{civ}} = 1$ in equation 5.2 and $\mu = 10$ in Eq. 5.10. An agent needing to push the garbage may lose up to 3 time-clicks, so an extra reward of $10 - 3 = 7$ will convince it to be ethical.
- In the over-regulated scenario reward function we set a punishment of $p_2 = 10$ for not complying with norm n_2 (see equation 5.3) for the same reasons we have set $\mu = 10$.

These parameters should theoretically allow the agents to learn an optimal and also ethically-aligned policy. By making $p_1 = 10$, we are guaranteeing that $R_N(s, a_*) = 0$ for every state s (where a_* is the action recommended by the optimal policy π_* in the state s), and similarly by making $\mu \cdot \text{eval}_{\text{civ}} = 10$, the action a_* will always be also the most ethically-aligned one.

6.3.2 Performance measures

We have defined six different performance measures. These measures are only applied when testing, namely after training ends up. The first two measures are related with the first hypothesis, the average time taken and the average tiredness:

- **Time taken:** measures the average of time-ticks each agent needs to get to its goal. Since agents are 3 cells away from their goal cells, we should expect a value always higher or equal than 3 and much smaller than 20, which is the maximum amount of time-ticks per episode
- **Tiredness:** average number of effective actions performed by agents to get to their goals (thus not counting dummy actions, actions with no effect on the environment). The range of values here should be very similar to the previous one (between 3 and 20, but ideally always very near to 3).

With them we can determine if the agents are being fast enough achieving their objectives and how much are they slacking off.

The following two measures are related with the second hypothesis, defined in terms of NF and EF measures respectively.

- **Violence:** informally, it measures the degree of harmfulness of the society the two agents live in. We can use the fact that norm n_1 exactly regulates that in order to formally define it as

$$\text{Violence}(\pi) \doteq 1 - \frac{\sum_{i=0}^m NF(\pi^i, \{n_1\})}{m}, \quad (6.1)$$

where NF is defined in eq. 4.6 and m represents the number of agents, which in our case is $m = 2$. With this formula we can know the average degree of norm-compliance of the agents. Since in practise this formula can be very hard to calculate, in our case we use the following estimation:

$$\text{Violence}(\pi) \approx \frac{\text{number of episodes an agent is hurt}}{\text{number of episodes of the simulation}}, \quad (6.2)$$

since violating the norm n_1 and hurting an agent are equivalent concepts in the public civility problem. With this formula, a violence rate of 1 indicates a very dangerous system, whereas a rate of 0 shows a peaceful and respectful community.

- **Tidiness:** it is independently measured for each agent, so we will consider as

$$Tidiness(\pi^i) \doteq EF(\pi^i, v_{civ}), \text{ with } in \in \{L, R\}, \quad (6.3)$$

where EF is defined in eq. 30. With this formula we can know if an agent's policy is promoting the moral value of civility and how much. As in the previous case, this can be hard to calculate. Thus, we will be using the following estimation:

$$Tidiness(\pi^i) \approx \frac{\text{number of times agent } i \text{ throws the garbage to the wastebasket}}{\text{number of times agent } i \text{ finds the garbage on its way}}. \quad (6.4)$$

A value of 1 means that an agent policy is promoting civility every time it can (always throwing the garbage to the wastebasket), whereas a value of 0 means the complete opposite.

The objective of the final two measures is to assess the global behaviour of the multi-agent system as a whole. These measures have been specifically designed for the public civility problem:

- **Semi-civility:** measures the number of episodes in which the garbage ends up being on a red cell (remember figure 5.1, in which red cells represent places that are not the wastebasket but that allow the agents to cross the street without further disturbances) divided by the total number of testing episodes. This measure gives us more information about the policy that the agents have learnt and allows us to know if they have learnt to simply put the garbage in a location where it does not bother them (a red cell).

we call this measure **semi-civility** because agents pushing the garbage to a red cell are not fully civil, but they are at least making the effort of clearing the street (without cleaning it though).

- **Civility:** measures the number of episodes in which the garbage ends up being on a wastebasket (i.e., green cell in Figure 5.1) divided by the total number of testing episodes. This measure gives us more information about the behaviour that the agents have learnt and allows us to know if they have overall learnt to properly dispose garbage.

Notice that the sum of **semi-civility** and **civility** do not necessarily sum 1, since there can be episodes in which the garbage is left in a passable grey cell. They are though slightly related and their sum can be at most 1.

6.3.3 Training

Since we have decided to use the framework of Markov games to model our research problems, we can also use reinforcement learning algorithms to train our agents.

Considering that our simulation is fully deterministic from a global perspective and relatively small (the number of state-action pairs is not exorbitant), we have decided to apply **Q-learning** to solve it. It is both easy to implement since it is a model-free off-policy algorithm, and also capable of finding an optimal solution under the right conditions. This method also serves as an initial attempt to tackle our problem, prior to trying more sophisticated algorithms in further research. Recall from section 3.3.1 that Q-learning uses the following updating formula (Eq. 3.15):

$$Q_{t+1}(s, a) = (1 - \alpha_t)Q_t(s, a) + \alpha_t \cdot (r + \gamma \max_{a' \in \mathcal{A}} Q_t(s', a')), \quad (6.5)$$

where the parameter α is the learning rate and the γ is the discount factor of the problem. We can change both parameters to control and improve the ways the agents learn.

For the single-agent case, the learning agent updates its conception of the world through this algorithm. Likewise, in the multi-agent case, each agent will have its own Q table being updated according to their own individual vision of the environment.

The simplicity of Q-learning allows us to be completely oblivious of the state transitions and also gives us a lot of freedom when selecting the training policy that the agents will follow while learning. In our simulation agents start knowing nothing about the environment, not even the most basic things such as knowing they should not move right where there is no physical path to the right, for that reason it will be very important that the training policy allows them to explore at least a handful of times every possible state-action pair.

In the following subsection we detail how we have set all the Q-learning parameters.

Q-learning parameters setup

In Q-learning there are not many parameters to set up. We have the discount factor γ , the learning rate α , the training policy and the number of training and testing episodes. Now we explain how we have chosen each of them.

- For the learning rate α and the discount factor γ we have applied **CRAN-IRACE** (Iterated Racing for automatic Algorithm Configuration) [26] in order to determine which are their optimal values for each scenario. IRACE, as its name implies, automates the tuning of the selected parameters of an algorithm by trying several configurations heuristically and testing them in order to find the most appropriate ones. We consider that this method is more sophisticated than the classical brute-force trial of all possible combinations.
- As a training policy we have decided to use the well-known ϵ -greedy policy. In our case we have defined an $\epsilon_t(s)$ dependent of the state the agents are, which decreases exploration for each state the more they visit it. The formula it uses is concretely:

$$\epsilon_t(s) \doteq \max(0.1, 0.99^{t(s)}), \quad (6.6)$$

where $t(s)$ corresponds to the number of times a state s has been visited. Notice that $0.99^t < 0.1$ if and only if $t \geq 230$. We believe that 200 opportunities per state are more than enough for the agents to discern the best action at each situation.

- The episodes are naturally defined as equivalent to days of our environment (remember that a day is the time lapse between the agents exiting their respective homes and arriving to their respective goals). We train a total of 3000 episodes or days in which every day can last a maximum of $Max_t = 20$ time ticks. Considering that both agents are only three cells behind their goal, we consider this episode limit to be reasonable. To speed up training time, once both agents arrive to their respective goals, we automatically advance to the next day. For a formal explanation, see the formula 5.5 of terminal states in our environment.

So in total we have 3000 episodes of a maximum of 20 time ticks each, which roughly represents $(3000 \cdot 20) / (560 \cdot 6) \simeq 18$ learning trials per state-action combination (but in fact as the training advances the agents will focus only in the most profitable state-actions).

- Finally, for the testing episodes we have decided to use 1/3 of the amount of training episodes. So we will be testing the learnt policies with 1000 episodes.

Notice that all the parameters will remain unchanged for all the scenarios. This is one of the strengths of Q-learning: its simplicity allows its use in many different environments with little parameter changes.

Benchmarking: a pre-programmed policy

In order to consider multi-agent scenarios populated by single learning agents, we define a pre-programmed policy to be used by the non-learning agents. Therefore, we use a stochastic policy which represents how a standard agent would act at each scenario. By default, an agent always moves forward towards its destination (goal position). Nevertheless, if a garbage blocks its way, it reacts differently according to the norms and ethical values established in the society of each scenario. Thus:

- In the *Unregulated* scenario, the agent simply pushes the garbage (i.e., performs pF , pL or pR) towards any randomly-chosen adjacent (existing) cell.
- In the *Regimented* scenario, the agent behaves as in the unregulated scenario but complying with norm n_1 , so that it avoids hurting the other agent by pushing the garbage towards random (but “safe”) positions.
- In the *Ethical* scenario the agent behaves *civilly* and pushes the garbage all the way to the wastebasket when it runs into it.
- In the *Over-regimented* scenario the agent behaves in the same fashion as in the ethical scenario.

Notice that this fixed policy includes a stochastic component that translates into an non-deterministic environment for the learning agent. This policy is a proof that ethically-aligned policies exist in the public civility problem, and therefore if all our theoretical results and setup are correct, the learning agent should be able to find it.

6.3.4 Testing

Here we enumerate the performed tests and in what consisted.

0. Before doing any actual experiment we have first conducted a pre-test in the 16 possible settings in order to determine the best possible learning parameters applying CRAN-IRACE, as we have mentioned in section 6.3.3.

After that we have done the actual testing. This testing always follows the same methodology for each of the 16 settings.

1. While the learning agent (or agents) train, we measure the evolution of the accumulated reward they obtain.
2. After the training episodes have finished, the learnt (and in the single-learning agent case, also the pre-programmed) policies are tested in a total of 1000 episodes. Exclusively from monitoring these 1000 episodes we calculate the ratios of the six measures we consider.

For the sake of avoiding the effects of randomness in the Q-learning algorithm (mainly due to the ϵ -greedy policy) we need to repeat each of these 16 tests several times. Concretely in our case we have repeated each test a total of 300 times. This means that each of these 300 times we have made the agent or agents re-train from scratch and then evaluated their performances in a new set of 1000 testing episodes.

3. The measures obtained each of these 300 times have been later averaged for each of the 16 settings. Furthermore, these measures have also served for statistical significance testing in order to asseverate the differences between settings and support our claims in the second hypothesis (to asseverate that the society does improve as a whole when its agents follow an ethically-aligned policy).

6.4 Results

Along this section we show the obtained results from the conducted experiments that we have explained in the previous section 6.3. In subsection 6.4.1 we show the pre-test results. Once the pre-test was done, we could train the agents as explained in subsection 6.3.3.

After the training was over and both agents had fixed policies we could perform the tests. In subsection 6.4.2 we detail the obtained results in terms of the six defined measures (including their statistical tests). Finally, in subsection 6.4.3 we show the results evaluated from a classical reinforcement learning point of view.

6.4.1 Pre-test results: CRAN-IRACE

As we have mentioned, as a pre-test we applied CRAN-IRACE in order to find the best parameters for the Q-learning algorithm in each setting, since it allowed us to use a more sophisticated methodology than simply brute-force trial and error of all possible combinations. We show the results first for the single-agent settings and then for the multi-agent settings.

In any case it is important to notice that this kind of methodology is not necessary nor is actually followed at all. Commonly the α and γ values are fixed and decided beforehand almost intuitively and it is highly probable that they will lead to a convergent policy.

Single-agent learning scenarios

Scenario	Variation	α	γ
Unregulated	Episodic	0.8	0.7
Regimented	Episodic	0.8	0.7
Ethical	Episodic	0.8	0.7
Over-regimented	Episodic	0.8	0.8
Unregulated	Sequential	1.0	0.8
Regimented	Sequential	1.0	0.8
Ethical	Sequential	1.0	0.8
Over-regimented	Sequential	1.0	0.8

Table 6.1: CRAN-IRACE recommended α and γ values for each scenario setting when there is only one learning agent.

In a single learning agent system, IRACE always considers very high α and γ values as we can see in table 6.1. They both make sense since a high α learning rate corresponds with a very deterministic learning process in which what has happened on a state will almost certainly happen again when the learning agent faces the same state. This is exactly the case here, since the other agent has a fixed policy depending on the scenario that never changes and which is only slightly stochastic (see subsection 6.3.3). The high discount factor γ makes sense since the episodes are finite.

Multi-agent learning scenarios

Scenario	Variation	α	γ
Unregulated	Episodic	0.5	0.8
Regimented	Episodic	0.5	0.8
Ethical	Episodic	0.6	0.7
Over-regimented	Episodic	0.6	0.8
Unregulated	Sequential	0.7	0.8
Regimented	Sequential	0.7	0.7
Ethical	Sequential	0.7	0.8
Over-regimented	Sequential	0.8	0.8

Table 6.2: CRAN-IRACE recommended α and γ values for each scenario setting when there are two learning agents.

We can see in table 6.2 that in the multi-agent agent case the learning rate α is smaller than the one for the single learning agent scenarios. These are results expected from the theory: the environment has become much more non-deterministic from the agents' point of view. We can see that the discount rate γ is very similar to the previous case. This again is also to be expected since we have not changed the episodes definition.

6.4.2 Evaluating performance measures

This section is devoted to present the results in terms of the measures previously introduced in subsection 6.3.2: for each of the 16 possible environment settings, we measured the 6 performance measures we were interested in. As in the previous subsection, we present first the results for single learning agents and then, the ones for two learning agents.

Single-agent learning scenarios

In this subsection we aim to analyse how a newcomer agent learns to behave in differently ethically evolved societies.

Firstly, we consider the episodic scenarios. Table 6.3 shows the results in terms of our first four performance measures. The first row shows that in the base-line *unregulated* scenario agents:

- (i) take an average time of 3.76 ticks per trip, which represents a 25% of increment compared to the 3 ticks required for reaching the goal position without the garbage blocking the way
- (ii) perform an average of 3.64 in terms of **Tiredness**, which is very similar to the **Time Taken** (implying that the number of dummy actions that agents take on average is very small) and
- (iii) the level of **Violence** is 68 per cent of the times, which indicates this is a wild, aggressive scenario.
- (iv) As for **Tidiness**, agent L learns to behave civilly 11% of times, whereas R with its pre-programmed policy does it 19% of the the times due to the randomness in its policy.

Scenario	Time Taken	Tiredness	Violence	L Tidiness	R Tidiness
Unregulated	3.76 ± 0.1	3.64 ± 0.04	0.68	0.11	0.19
Regimented	4.1 ± 0.03	3.81 ± 0.01	0.0	0.33	0.41
Ethical	4.13 ± 0.09	4.0 ± 0.01	0.0	1.0	1.0
Over-regimented	4.19 ± 0.1	4.0 ± 0.01	0.0	1.0	1.0

Table 6.3: Results when testing with episodic settings in terms of the first four measures. Learning agent (L); Non-learning agent (R).

Scenario	Semi-civility	Civility
Unregulated	0.15 ± 0.01	0.15 ± 0.01
Regimented	0.37 ± 0.02	0.37 ± 0.02
Ethical	0.0 ± 0.0	1.0 ± 0.0
Over-regimented	0.0 ± 0.0	1.0 ± 0.0

Table 6.4: Results when testing with episodic settings in terms of the global measures.

Subsequent rows in Table 6.3 show a different phenomenon for alternative scenarios. The *regimented* scenario tackles the undesirably high aggressiveness in the unregulated scenario by enacting norm n_1 , which bans such behaviour. The side effects of reducing **Violence** down to 0 are two-fold. First, **Time Taken** increases almost a 10%, although **Tiredness** does not increase as much, because not hurting the other agent can be effortlessly done by simply waiting for the other agent to move to another (safer) cell. Second, **Tidiness** increases because agents sometimes throw the garbage to the wastebasket (the rest of times they leave it aside into the red cells). Agent L (with **L Tidiness**=0.33) learns this norm-compliant behaviour in order to avoid the punishment of norm n_1 , whereas the pre-programmed policy of agent R changes its probabilities of pushing the garbage (see subsection 6.3.3). Table 6.4 provides some insights about the cells where the garbage ends up at the end of the episodes. Thus, on average, for the *unregulated* scenario, the garbage ends up on a grey cell (i.e., blocking the way) 70% of the times, and the remaining 30% is equally distributed among red and green (wastebasket) cells. Now, considering the *regimented* scenario, the garbage ends up blocking the way far less times (26%) but it stills distributes the garbage equally between red and green cells (37% each).

As for the *ethical* scenario in which norm n_1 is enacted and the moral value v_{civ} is promoted, it does not only keep **Violence** down to 0, but also increases **Tidiness** up to 1 by always throwing the garbage to the wastebasket (columns 2 and 3 in Table 6.4 indicate the garbage ends up on the green cells 100% of the episodes). Obviously, there is a price to pay, since both **Time Taken** and **Tiredness** have a slight increase (specially notable in the case of **Tiredness**). The reason is the average extra time and effort agents take to tidy up the street. Thus, agent L learns to sacrifice part of its individual goal of reaching its goal as fast as possible to avoid aggressive behaviours and dirty streets, showing a praiseworthy behaviour.

To conclude the episodic part, the *over-regimented* scenario shows results extremely similar to the previous one, with the only difference being a small increase of less than 2 % in **Time Taken**, which probably is due to some statistical error. These results are not surprising since we are encouraging the agents to enact the same behaviours as in the *ethical* scenario.

Secondly, we consider the sequential variations (with sequences of $d = 10$ episodes) for the four alternative scenarios. Table 6.5, together with table 6.6, show our results.

Scenario	Time Taken	Tiredness	Violence	L Tidiness	R Tidiness
Unregulated	3.32 ± 0.05	3.23 ± 0.02	0.23	0.12	0.18
Regimented	3.2 ± 0.04	3.12 ± 0.01	0.0	0.33	0.44
Ethical	3.14 ± 0.05	3.1 ± 0.0	0.0	1.0	1.0
Over-regimented	3.14 ± 0.05	3.1 ± 0.0	0.0	1.0	1.0

Table 6.5: Results when testing with sequential settings in terms of the first four measures.

Scenario	Semi-civility	Civility
Unregulated	0.38 ± 0.04	0.39 ± 0.05
Regimented	0.48 ± 0.06	0.49 ± 0.06
Ethical	0.0 ± 0.0	1.0 ± 0.0
Over-regimented	0.0 ± 0.0	1.0 ± 0.0

Table 6.6: Results when testing settings in terms of the global measures.

Interestingly enough, the average results in all four scenarios are pretty similar in terms of **Time Taken** and **Tiredness** (from 3.14 to 3.32 in Table 6.5). This decrease happens because agent L learns to unblock the way pushing the garbage to a red or green cell. That way, the agent can seamlessly reach its goal destination as fast as possible the remaining episodes in the sequence until the garbage changes its location again at the end of the d -th episode. In this case we are going to compare each measure result with its equivalent in the episodic scenarios.

In the *unregulated* scenario, 38% of the times the garbage ends up being in red (dumpy) cells, and another 39% it is located on green (wastebasket) cells (see table 6.6). Compare this with the values in the episodic *unregulated* scenario, in which 70 % of the episodes the garbage ended up on a grey cell. However we can see that in terms of **Tidiness** nothing has changed much (from 0.11 and 0.19 to 0.12 and 0.18 respectively). This is related to what we have just explained: Tidiness does not increase when the agent pushes the garbage to a red cell (in fact it decreases). **Violence** on the other hand, has been drastically reduced (from 68 % to 23 %) as an additional side effect of the new agent policy. Overall, we find this result on improved performance of particular interest because it indicates that implicit ethical issues naturally emerge on this sequential scenario because they also favour system efficiency. Thus, we can think of norms and moral values as mechanisms implicitly present when improving the long-term coordination in the society. In other words, these results seem to support the idea that moral values are socially created mechanisms that capture performance requirements. This can be intuitively thought as “if agents behave right (civically), things will eventually go better than if they behave selfishly”.

Likewise the episodic scenario, explicitly considering the moral value further improves performance. In fact, the moral value proves to be so effective in coordinating the society that agents in the *ethical* scenario even take slightly less average time than those in the *unregulated* scenario (see **Time Taken** in Table 6.5, going down from 3.32 to 3.14).

This phenomenon is again reproduced when considering the *over-regimented* scenario: **Time Taken** and **Tiredness** are almost minimal (remember that their minimal value is 3) and both agents act in an impeccably civic way.

Multi-agent learning scenarios

In this subsection we aim to analyse how emerge differently ethically evolved societies when considering more than one learning agent. It is important to remark that this is a harsher environment for the agents to learn since now they are both learning and changing their policies.

Likewise the previous subsection, we firstly consider the episodic scenarios. Table 6.7 shows the results in terms of the first four performance measures. The results are very similar to the ones we can

appreciate in table 6.3, when only one agent was learning its policy. The first row shows that in the base-line *unregulated* scenario the R agent now has a Tidiness of 0.08 (which represents a reduction of 80 %) while the L agent has a very similar value. This decrease explains why in the first row of table 6.8 both the degrees of *Semi-civility* and *Civility* have decreased.

Scenario	Time Taken	Tiredness	Violence	L Tidiness	R Tidiness
Unregulated	3.77 ± 0.1	3.67 ± 0.06	0.68	0.08	0.08
Regimented	4.08 ± 0.08	3.93 ± 0.05	0.0	0.45	0.4
Ethical	4.13 ± 0.09	4.0 ± 0.01	0.0	1.0	1.0
Over-regimented	4.15 ± 0.11	4.0 ± 0.01	0.0	1.0	1.0

Table 6.7: Results when testing with episodic settings in terms of the first four measures..

Scenario	Semi-civility	Civility
Unregulated	0.08 ± 0.06	0.08 ± 0.06
Regimented	0.42 ± 0.11	0.43 ± 0.12
Ethical	0.0 ± 0.0	1.0 ± 0.0
Over-regimented	0.0 ± 0.0	1.0 ± 0.0

Table 6.8: Results when testing with episodic settings in terms of the global measures.

In the *regimented* scenario the most notable difference this time is that the L Tidiness has increased from 0.33 to 0.45. This might seem odd since the second row of table 6.8 shows in fact a decrease from 0.49 to 0.43 in the green cells percentage, but this is only apparent since the reduction in the mean is compensated with an increase of the standard deviation from 0.06 to 0.12. Therefore, this change is not so significant after all.

The *ethical* and *over-regimented* scenarios show results almost indistinguishable between them and their single learning agent counterparts.

Now we consider the sequential variations for the four alternative scenarios with two learning agents. Table 6.9 together with table 6.10 show the results.

Scenario	Time Taken	Tiredness	Violence	L Tidiness	R Tidiness
Unregulated	3.32 ± 0.58	3.16 ± 0.15	0.16	0.3	0.26
Regimented	3.17 ± 0.35	3.1 ± 0.05	0.0	0.49	0.46
Ethical	3.13 ± 0.12	3.1 ± 0.02	0.0	0.99	0.99
Over-regimented	3.18 ± 0.36	3.09 ± 0.07	0.0	0.99	0.99

Table 6.9: Results when testing with sequential settings in terms of the first four measures.

Scenario	Semi-civility	Civility
Unregulated	0.42 ± 0.2	0.41 ± 0.2
Regimented	0.48 ± 0.16	0.51 ± 0.16
Ethical	0.0 ± 0.0	1.0 ± 0.0
Over-regimented	0.0 ± 0.0	1.0 ± 0.0

Table 6.10: Results when testing with sequential settings in terms of the global performance measures.

Again the average results in all four scenarios are pretty similar in terms of *Time Taken* and *Tiredness* and also very similar to the ones we already showed in table 6.5.

It is interesting to remark that this time *Tidiness* in both the *ethical* and the *over-regimented* scenarios is not 1 but 0.99, which might be related to the fact that this is the hardest environment for

the agents to learn civility. Thus, sometimes (very rarely) Q-learning is not enough for them to learn the best possible policy. In any case, in the last two rows of table 6.10 we can see that the garbage ends up on a green cell every episode. This can be explained if we consider that the agents very rarely learn a strange behaviour that we like to call *free riding*: one of the agent pushes the garbage to the side and the other learns to prevent this and waits for that to happen so it can push the garbage to the wastebasket.

Overall we observe that by simply using Q-learning on each agent individually they have been able to learn policies almost identical as the ones that the L agent learnt in the single learning agent settings. This was to be expected since the public civility problem was a simple environment after all, but we will probably need to considered algorithms specialised for multi-agent reinforcement learning if we plan to create a more complex environment to study civility. This issue, as we have seen, is already highlighted when we consider the setting with 2 learning agents in sequential ethical or over-regimented scenarios.

Statistical analysis

In this subsection we provide statistical support for the claims we have made in the previous two subsections. Concretely we show statistical tests that support these two claims:

- Policies learnt in *ethical* scenarios are much more civic than the ones in the unregulated and regimented scenarios.
- Policies learnt in *ethical* and *over-regimented* scenarios are very similar between one another.

In order to perform the actual testing we first needed to do a normality test applied to each setting and individual measure in order to check which was the most appropriate set of distribution tests for each case. We have thus performed a total of $5 \cdot 16 = 80$ **Kolmogorov-Smirnov** tests (where 5 represent the number of individual measures, counting Tidiness twice, and 16 are the number of scenario settings) . Prior to that, we have first cleaned the populations from outliers (via calculating their respective box plot diagrams and discarding the data points outside the whiskers), that would otherwise perturb too much the test results. Tables 6.11, 6.12, 6.13 and 6.14 show the obtained normality test results in detail. To sum up, all the 80 tests rejected the null hypothesis of normality (i.e., it none of the 80 populations follow a normal distribution).

Scenario	Time Taken		Tiredness		Violence		L Tidiness		R Tidiness	
	D	p	D	p	D	p	D	p	D	p
Unregulated	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.727	$< 10^{-3}$	0.53	$< 10^{-3}$	0.557	$< 10^{-3}$
Regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.588	$< 10^{-3}$	0.635	$< 10^{-3}$
Ethical	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.841	$< 10^{-3}$	0.841	$< 10^{-3}$
Over-regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.841	$< 10^{-3}$	0.841	$< 10^{-3}$

Table 6.11: Kolmogorov-Smirnov results for SARL episodic settings.

Scenario	Time Taken		Tiredness		Violence		L Tidiness		R Tidiness	
	D	p	D	p	D	p	D	p	D	p
Unregulated	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.556	$< 10^{-3}$	0.509	$< 10^{-3}$	0.541	$< 10^{-3}$
Regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.526	$< 10^{-3}$	0.608	$< 10^{-3}$
Ethical	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.838	$< 10^{-3}$	0.841	$< 10^{-3}$
Over-regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.841	$< 10^{-3}$	0.841	$< 10^{-3}$

Table 6.12: Kolmogorov-Smirnov results for SARL sequential settings.

Scenario	Time Taken		Tiredness		Violence		L Tidiness		R Tidiness	
	D	p	D	p	D	p	D	p	D	p
Unregulated	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.652	$< 10^{-3}$	0.5	$< 10^{-3}$	0.5	$< 10^{-3}$
Regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.588	$< 10^{-3}$	0.506	$< 10^{-3}$
Ethical	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.841	$< 10^{-3}$	0.838	$< 10^{-3}$
Over-regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.841	$< 10^{-3}$	0.841	$< 10^{-3}$

Table 6.13: Kolmogorov-Smirnov results for MARL episodic settings.

Scenario	Time Taken		Tiredness		Violence		L Tidiness		R Tidiness	
	D	p	D	p	D	p	D	p	D	p
Unregulated	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.5	$< 10^{-3}$	0.5	$< 10^{-3}$
Regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.5	$< 10^{-3}$	0.5	$< 10^{-3}$
Ethical	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.831	$< 10^{-3}$	0.808	$< 10^{-3}$
Over-regimented	1.0	$< 10^{-3}$	1.0	$< 10^{-3}$	0.5	$< 10^{-3}$	0.835	$< 10^{-3}$	0.835	$< 10^{-3}$

Table 6.14: Kolmogorov-Smirnov results for MARL sequential settings.

With these results we have proceeded to perform the non-parametric tests (concretely the **Kruskal-Wallis test**) in order to show from a statistical point of view the decrease in **Damage** in the normative scenarios and the increase in **Tidiness** in the ethical scenarios. In this case this was not needed since the increase was really notorious, but we have decided to do the tests in any case for the sake of being methodical. The tests are only performed from the results of the L agent. Tables 6.15 and 6.17 shows the results in the SARL settings whereas tables 6.16 and 6.18 shows the respective results in MARL settings.

These tables are all organised as follows: in the first row they present the test results when comparing episodic scenarios and in the second one when comparing sequential scenarios. The first two columns are the test results when comparing the ethical versus the unregulated scenarios, and similarly the last two columns are the results when comparing the ethical versus the regimented scenarios.

	Ethical v. Unregulated		Ethical v. Regimented	
	K	p	K	p
Episodic	513.45	$< 10^{-3}$	0.0	1.0
Sequential	513.19	$< 10^{-3}$	0.0	1.0

Table 6.15: Kruskal-Wallis test results when comparing **Violence** in SARL settings. In the first row it presents the test results when comparing episodic scenarios and in the second one when comparing sequential scenarios. The first two columns are the test results when comparing the ethical versus the unregulated scenarios, and similarly the last two columns are the results when comparing the ethical versus the regimented scenarios.

	Ethical v. Unregulated		Ethical v. Regimented	
	K	p	K	p
Episodic	480.34	$< 10^{-3}$	0.0	1.0
Sequential	412.86	$< 10^{-3}$	0.0	1.0

Table 6.16: Kruskal-Wallis test results when comparing **Violence** in MARL settings.

The results in tables 6.15 and 6.16 confirm without doubt that the violence decrease in the normative scenarios is significant and that it does not improve in the ethical scenario (since it is already null).

	Ethical v. Unregulated		Ethical v. Regimented	
	K	p	K	p
	513.43	$< 10^{-3}$	512.70	$< 10^{-3}$
	513.43	$< 10^{-3}$	512.70	$< 10^{-3}$

Table 6.17: Kruskal-Wallis test results when comparing L Tidiness in SARL settings.

	Ethical v. Unregulated		Ethical v. Regimented	
	K	p	K	p
	520.99	$< 10^{-3}$	499.81	$< 10^{-3}$
	521.27	$< 10^{-3}$	492.46	$< 10^{-3}$

Table 6.18: Kruskal-Wallis test results when comparing L Tidiness in MARL settings.

The results in tables 6.17 and 6.18 confirm without doubt that the tidiness increase in the normative scenarios is significant and that the increase in the ethical scenarios respect the normative scenarios is also significant.

After that we have then tested the similarities between the learnt policies in ethical and over-regimented scenarios. Table 6.19 shows the results we have found. You will notice that the results in terms of Tidiness are almost too good to be true. This is because we have eliminated several outliers which were producing very perturbed results, as we have mentioned before. So we can conclude that the ethical policies in the SARL settings are almost identical to the over-regimented policies and also identical to the policies they learn in the MARL settings, which was even a stronger fact than our initial claim.

Variation	Time Taken		Tiredness		Violence		L Tidiness		R Tidiness	
	K	p	K	p	K	p	K	p	K	p
Episodic	59.845	$< 10^{-3}$	0.538	0.910	0.0	1.0	0.0	1.0	0.0	1.0
Sequential	46.996	$< 10^{-3}$	7.274	0.064	0.0	1.0	0.0	1.0	0.0	1.0

Table 6.19: Kruskal-Wallis test results on Ethical versus Over-regimented scenarios. This test compares together SARL and MARL settings.

6.4.3 Q-values analysis

In this subsection we show the evaluation results from a more classical reinforcement learning perspective. Without lose of generality all results here only refer to the L agent, which are extremely similar to the R agent results. Firstly we show how the reward function of the L agent evolved while training and then we show the actual policy it learnt. For the illustrations and graphics we have selected for each setting the results of an agent that learns the mode of the policies in each setting.

Accumulated reward

All the figures in this subsection show the averaged accumulated reward that the L agent obtains per episode. The accumulated reward is the sum of all the rewards the agent obtains in an episode, so in this case, if we take for instance the unregulated scenario, the maximum value of the accumulated reward is $Max_t - 3 = 17$, where $Max_t = 20$ and 3 represents the 3 cells that agents need to cross to get to their respective goal position. For the sake of reducing the noise produced by the randomness while training, we have averaged these accumulated rewards the following way: given an episode we represent the average accumulated reward obtained in previous 100 episodes.

Figure 6.1 shows the results obtained with the single learning agent settings. The unregulated (blue) curve serves as the baseline curve in both the episodic and sequential scenarios. We can

appreciate that they follow a very similar trajectory: they start at -20 (the lowest possible value, meaning that the agent is incapable of getting to the goal position) and quickly these values rise in less than 300 episodes up to 10 (meaning that it knows how to get to the goal position but still the ϵ that determines when to act randomly is so high that it still does not follow the optimal route). We observe that in 500 episodes they both stabilise at around 15. All the rest of the curves are simple deviations from these two, evidencing that the rest of scenarios are extensions of the unregulated one.

The regimented (orange) curve is almost equal to the unregulated one, except that it sometimes has a lower value due to norm violations. We can see that at the end this difference is hard to detect, which means that the agent has learnt to comply with n_1 (see Eq. 5.1), the norm in place.

The ethical (green) curve is always the one that grows the most (getting to up to 20 in the episodic scenarios), which was to be expected since only in the ethical scenario the reward function gives an extra positive reward associated with throwing the garbage to the wastebasket. Concretely the maximum reward it can get is $(Max_t + \mu) - (3 + 3) = (20 + 10) - 6 = 24$. Considering that on average the agent only gets the μ surplus half of the times (when the wastebasket is on its side) its reward should stabilise at around $(24 + 17)/2 = 20.5$ which is exactly what it does. This surplus however is hardly noticeable in the sequential scenarios (the graphic on the right), since the agent just has the opportunity to throw the garbage at the wastebasket once every ten episodes. In any case this is indicating us that the agent has both learnt to throw the garbage at the wastebasket (to behave ethically) and also an optimal policy from its point of view.

The over-regimented (red) curve on the other hand is the one that always has the lowest values. This is strongly related with the fact that it is the scenario with the most punishing reward function derived from both norms n_1 and n_2 in equations 5.1 and 5.3 respectively. As in previous cases, since the curve at the end is very similar to the unregulated one, we can conclude that yet again the agent has learnt to follow all the established regulations.

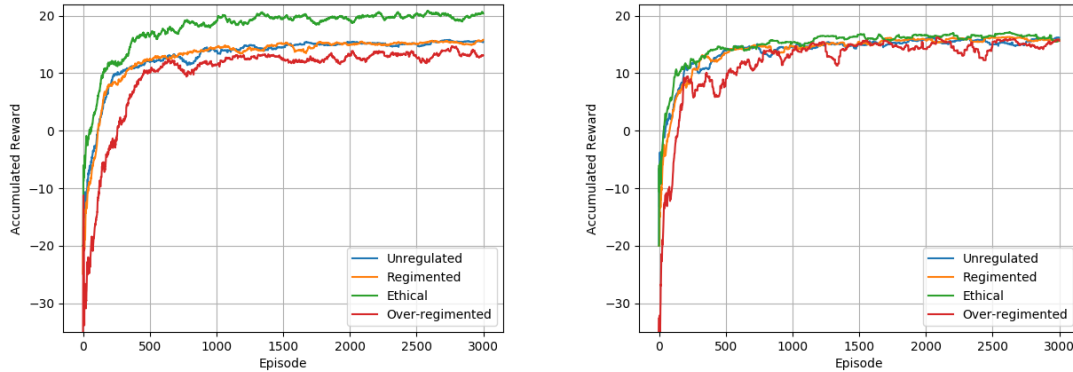


Figure 6.1: Evolution of the accumulated reward with the SARL settings. Left: episodic scenarios. Right: sequential scenarios.

As for figure 6.2, it shows the results obtained for the two learning agents settings. In general, they all follow the same patterns that we have seen before with the single learning agent settings (being the ethical scenario the one accumulating the most rewards). The most notable difference here is that all the curves seem to be spikier or noisier than their counterparts with a single learning agent. This is related with the problem of now having to predict also the erratic behaviour of the other learning agent, whose policy also changes as both learn.

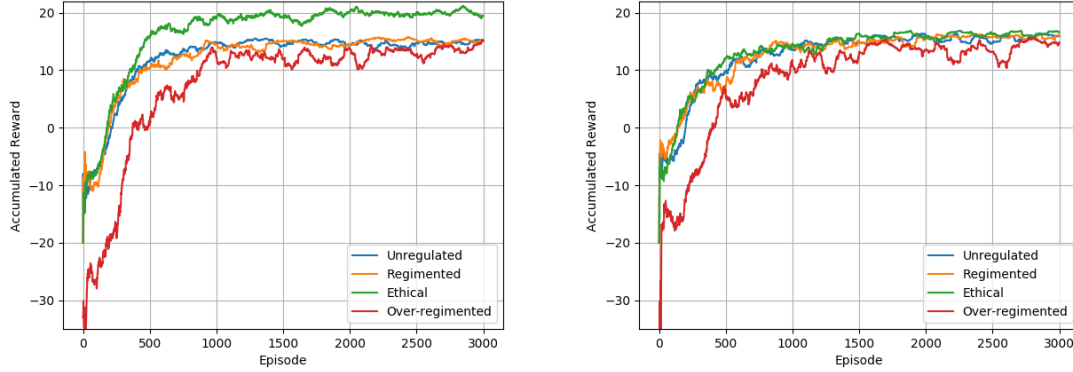


Figure 6.2: Evolution of the accumulated reward with the MARL settings. Left: episodic scenarios. Right: sequential scenarios.

The general conclusion we can extract from all these graphics (and particularly the ones from the ethical scenario) is that the agents always manage to learn an optimal/Nash equilibrium policy (that is also ethically-aligned in the ethical scenario), since in both situations all the curves stabilise at the expected reward values. Also, observing the shapes of the curves and how fast most of them tend to stabilise, we conclude that in some of them we have made an excessive training of many more episodes than the ones the agents really needed. This can be appreciated specially in the left graphic of figure 6.1, because all the curves stabilise at around 1000 training episodes.

Learnt policies

For this subsection we take a different approach to the previous one and we will briefly characterise the mode (most frequent) of the policies that the L agent learns in each scenario. This is more informative than showing the graphics for the accumulated rewards with the following policies since it would add no extra information to what we have explained in the previous subsection.

There are no differences between the mode (most frequent) of the policies learnt between the single-learning-agent and two-learning-agent settings. Therefore, we only need to characterise six policies: three for the episodic scenarios and three for the sequential scenarios. Notice that the policy learnt for the *ethical* and *over-regimented* scenarios is the same.

Figure 6.3 illustrates the learnt policy for the *ethical* and *over-regimented* scenarios since it is an easy and understandable example to show. In the figure, arrows indicate the direction of the action and the colour inside notates either if the action is **to move** (in solid black) or **to push** (colourless).

The agent always behaves the same way when there is no obstacle in front of it: it moves forward (as we can see from the left image of figure 6.3). The differences only arise when they are faced with the decision of what to do with the garbage that is blocking its way. In the **episodic** scenarios it acts the following way:

- In the *unregulated* scenario, it pushes to the right the garbage obstacle so it can free its path and get to the goal as soon as possible (without caring about the other agent's position). In the cases when the garbage is located in the goal cell, it may happen that the agent instead pushes it to the left or up, since it has an equivalent outcome from the agent's point of view.
- In the *regimented* scenario, the agent first looks if the other agent is nearby (in the adjacent cell to the right of the L agent or of the garbage). If this is the case, the agent waits doing a dummy (ineffective) action or pushing up the garbage. If not, it again pushes to the right the garbage. In the cases when the garbage is located in the goal cell, it may also happen that the agent pushes it up or left, since it has an equivalent outcome from the agent's point of view.

- In the *ethical* and *over-regimented* scenario it always pushes forward the garbage until it is adjacent to the wastebasket. In that moment the agent proceeds to push left the garbage so the obstacle gets inside the wastebasket (the green cell). Figure 6.3 illustrates this behaviour.

Therefore, we see that the learnt policies are quite similar to the pre-programmed policy we developed initially.

Furthermore, in the **sequential** scenarios the agent changes its policy only in the *unregulated* and *regimented* scenarios, becoming more complex. The change is in the fact that the agent becomes much more **semi-civic** on average (see subsection 6.3.2), specially in the regimented scenario. This is because the agent most of the times gets to the conclusion that this policy allows it to get faster to its goal for the rest of the episodes.

Finally, to watch the agent following this and any of the two other policies in either episodic or sequential scenarios we refer to the following video links, in which we monitor the agents' behaviours for 30 episodes in each video:

- Unregulated policy in an episodic scenario: <https://youtu.be/ibBdDutBddk>.
- Regimented policy in an episodic scenario: https://youtu.be/5etr_vv8QgQ
- Ethical/Over-regimented policy in an episodic scenario: <https://youtu.be/g37v6ICdiyM>
- Unregulated policy in a sequential scenario: <https://youtu.be/ck-aUwBpBEU>.
- Regimented policy in a sequential scenario: https://youtu.be/0boB00aJ6_4
- Ethical/Over-regimented policy in a sequential scenario: <https://youtu.be/o09x2SfJWOM>

At the bottom up of every video you will see the results for several measures (Time Taken, Tiredness, Violence –called Damage in the videos–, Civility) at each episode.

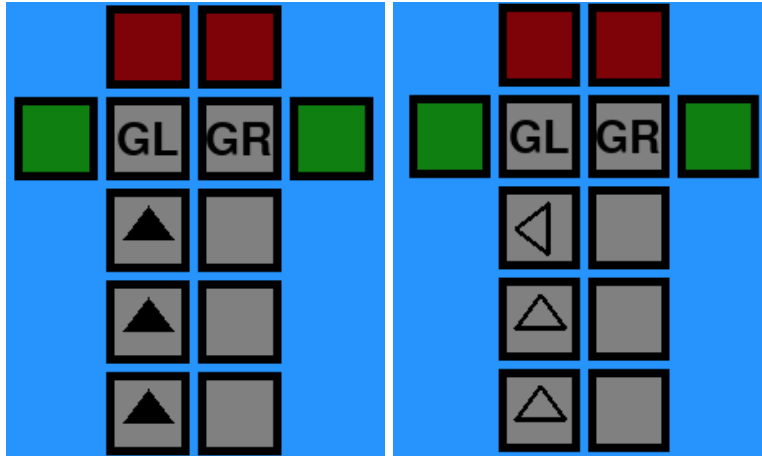


Figure 6.3: Policy the L agent learns in the ethical and over-regimented scenarios. Left picture represents the policy when there is no obstacle in front of the agent, whereas the right one represents the policy when there is an obstacle in front of the agent.

Chapter 7

Conclusions and future work

This chapter dissects the conclusions derived from this master thesis. Section 7.1 summarises the contributions that this thesis presents, explaining how they answer our three research questions in the introductory chapter. Then, section 7.2 presents some paths for future work.

7.1 Conclusions

This master thesis has successfully answered our research questions regarding moral values definition, ethical learning, and benefit assessment as posed in section 1.2. In this section we recap how we tackle them by summarising our main **contributions** :

Moral values formalisation

The first research question of this thesis proposed to tackle formal structure of a moral value. We answered this question in section 4.3 of Chapter 4, by explicitly formalising the concept of moral value.

The structure of a moral value, as formalised in definition 17, is a tuple $v = \langle \mathcal{N}_v, f_v \rangle$ consisting of two elements. The first one being the set of norms $\mathcal{N}_v$ that support it (in accordance with the deontic sphere of moral philosophy), that describes the minimal conditions of morality. And the second one being a recommendation function f_v that signals how good actions are with respect to that moral value (in accordance with the axiological sphere of moral philosophy), which allows us to describe the maximal conditions of morality.

Ethical learning framework

The second research question proposed to tackle the learning of ethical behaviours. This question is addressed in section 4.4 of Chapter 4 by defining Ethical Markov games.

Ethical Markov Games are Markov Games associated with a moral value $v = \langle \mathcal{N}_v, f_v \rangle$. They are extensions of classic Markov games whose reward function is extended with an ethical function (associated with f_v) that promotes actions aligned with the moral value via positive rewards, and a normative function that penalises actions that violate any norm of the set $\mathcal{N}_v$ via negative rewards.

With Ethical Markov Games we can change the objective policy of an agent towards a new and ethically-aligned one. They are the *tool*, but we still need the *instructions*, which are given in the next contribution.

Characterisation of ethically-aligned policies

In order to guarantee that the reward extension of the Ethical Markov games makes the agents learn behaviours aligned with a moral value, a careful design of the reward function is needed. In theorem

6 of Chapter 4 we provide the necessary and sufficient conditions that the reward function of each agent has to fulfil in order to ensure that it learns an ethically-aligned policy. Namely, it needs to compensate the long-term losses of behaving ethically and make violating norms unappealing. This is done by:

- (i) receiving a good enough positive reward for performing actions recommended by f_v and
- (ii) having penalties associated with the norms in $\mathcal{N}_v$ big enough to discourage violation them in any situation.

This way we can ensure that an agent that learns an ethically-aligned policy will comply with all the norms and perform praiseworthy actions every time it can.

Assessment of the social benefits derived from ethical behaviours

The third research question the quantification of the social benefits derived from the ethically-aligned behaviours of agents. We answer this last question by presenting, solving and evaluating the formal example of the *Public Civility Problem*, that we formalise as a Markov game in chapter 5. There we define the problem of learning to behave ethically in an urban context, where garbage should be properly thrown to the wastebasket. In it agents are equipped with RL algorithms that learn behaviours aligned with the *civility* moral value.

In chapter 6 we have explained how we have solved the Public Civility Problem. We have extended the reward functions of the agents with ethical and normative components. The values of the rewards have been carefully set according to theorem 6) in order to ensure that the agents learn ethically-aligned policies.

Afterwards, we have performed several tests to their learnt policies in order to assess the social benefits they provided to society. During the simulations, data of generic and specific metrics have been collected (e.g. degree of violence, tidiness). In addition, we have measured the degree of civility of the street, by counting the number of episodes that the garbage ends up in the wastebasket, in order to check the achievement of social welfare.

The empirical results show that the agents only learn an ethical behaviour when we add an ethical component to their reward function, validating the theoretical results from our theorems. Consequently, the increase of civility of the society is unquestionable with respect to when the agents use the baseline reward function. In fact, the society obtains the maximum possible degree of civility when the agents follow an ethically-aligned policy.

7.2 Future work

While this thesis has aimed to contribute to the ethical learning problem, its results also open new unexplored routes for future work. In this section, we present two problems where there is still a need for future research.

Moral values coexistence

It is still an open question how should we define the reward function of a learning agent when it has to be aligned with two or more moral values. In this master thesis we have only explored the case in which its policy has to align with one moral value, but probably some of our theoretical results must need to be expanded for allowing an agent to fulfil to several moral values, especially in situations in which they give contradictory recommendations or regulations.

Considering norms that are independent of moral values

A similar problem that remains unsolved is the approach that should be taken when an agent needs to learn a moral value and also an independent set of norms. It is unclear what the agent should choose in the cases in which the moral value recommends an action that is also forbidden by a norm.

Further testing our ethical learning framework

We have tested our framework in the Public Civility Problem, and we have seen that by using Q-learning we obtain near-optimal results. But it would be interesting to test it as well in other more complex scenarios (specially with more learning agents), to see whether with classical RL algorithms the agents can still manage to obtain Nash equilibria ethically-aligned policies or if we need to develop new algorithms specially meant for Ethical Markov games. In particular we could consider a generalisation of the Public Civility Problem with $m \gg 2$ agents.

We would also be interested in considering the Public Civility Problem with a new ethical function that recommends more actions. For instance, pushing the garbage to a red cell could be recommended with a value of 0.5.

Chapter 8

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